

Cognitive Biology: The Human Model and the GWAS Scaffold

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Abstract

The companion mouse-model paper builds a mouse embryo from a genome. Its compiler is a neural cellular automaton whose per-cell rule is read by an outer genomic model through four morphogenetic attention heads, validated against the whole-embryo Stereo-seq atlas. This paper carries that construction to the human. The architecture is a stack of four layers, each already seeded. A frozen *kernel*, the mouse compiler, supplies the conserved mechanisms of vertebrate body-building. A low-rank *species adapter*, a small move on the coupled attention-head manifold tuned against the human Stereo-seq embryo, carries the kernel to a human embryo. A *developmental forward* layer matures the embryo toward the adult. And a low-rank *individual adapter*, the genome-wide association effect sizes written as a bias on the outer model, carries the variation between people. The method is one chain, and every figure is an instance of it. The model's units are *attention heads*: master-transcription-factor super-enhancer clusters read from the genome by a regulatory front-end assembled from ABC enhancer-promoter contacts, the SEDb super-enhancer atlas and the AlphaGenome sequence model. The heads are grown into a body, and each is held to account by triangulation across three independent atlases: the dense human Stereo-seq embryo reconstruction (HESTA) for developmental shape, *Gray's* anatomy for named form, and the Physiome and BETSE models for function. The first result is that the kernel transfers. A human embryo is recovered from the mouse compiler with only a low-rank species tweak. Its cardiac loop scores against a dense human reconstruction as well as against the mouse, and the one organ the kernel misplaces, the gut, is the one the human lineage most remade. The central idea is the layer that closes the stack into a loop. Because the embryo is the cause of the adult, an adult association signal, where human data are abundant, is an indirect assay of the embryonic regulation that built the trait, where data are scarce. Read backward through the forward model, an effect size at a known locus tells the model which embryonic parameter it sets and in which direction, so the data-rich adult supervises the data-poor embryo. The same partition separates the timing of gene action, assigning an adult effect to the embryonic layer or to the maturation layer. We run the loop end to end on real association data. It closes to the embryo for the heart, eye and kidney, and the partition assigns the liver signal to maturation; we then extend it to the face and the whole body. The individual adapter is completed from the whole GWAS Catalog. Asked which genes reach many heads and which reach one, the catalog partitions the genome into

the conserved kernel and the individual adapter unprompted, leaving as kernel the embryonic-transient heads that no adult trait can vary. And because the whole construction hangs from one telomere-gated clock, that clock runs both ways. Wound forward it grows the human body; wound backward it returns the body, through the collapse of its electric axes, to the blastocyst and the single cell.

1 From the Mouse Model to a Human

The companion paper on the mouse model takes the cell-resolution compiler, a neural cellular automaton whose per-cell rule is read from the genome by an outer large genomic model through four morphogenetic attention heads (Figure 1), to the real mouse, and builds it out against the whole-embryo Stereo-seq atlas of organogenesis [6, 4, 5]. Grown forward from a single cell the compiler produces a body a reader recognises as a vertebrate, places its organs at their real antero-posterior levels, and resolves its head set in the order the clock predicts. Given a dense three-dimensional reconstruction of the fetus, it takes the right internal form, the cardiac loop and the coiled gut among them [3]. The object of this paper is the human. The claim is that the same compiler reaches it, not by being rebuilt, but by being adapted in a small number of low-rank steps and then extended forward in developmental time.

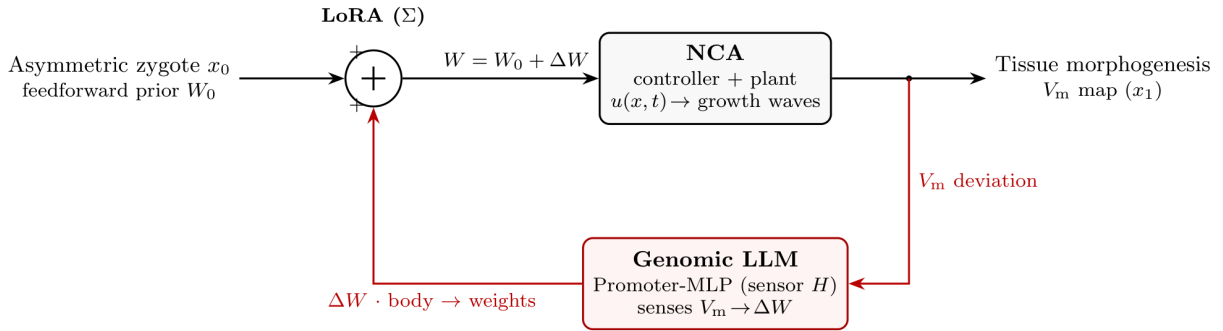


Figure 1: The NCA+LGM control architecture of the series, in standard closed-loop form, reproduced from the foundational paper [1]. The reference input, the asymmetric zygote x_0 , enters as the feedforward prior W_0 at the summing junction Σ ; the forward path is the inner neural cellular automaton (the Tissue Recursive Model, controller-plus-plant), whose vector field $u(x, t)$ integrates cell state to the output, the tissue V_m map x_1 . A pickoff feeds the feedback element H , the frozen Large Genomic Model, a Promoter-MLP sensor, which reads the V_m deviation and returns a low-rank correction ΔW into Σ : the summing junction is the LoRA, $W = W_0 + \Delta W$. The feedforward path already computes the recognisable phylotypic form; the feedback branch supplies the species- and individual-specific remainder. The species adapter and the GWAS individual adapter of this paper are two settings of that feedback branch ΔW .

The reason the human is hard is not the same at both ends of development. The human *embryo* is where data are scarce and, for the regulatory assays the compiler most wants, ethically restricted. The human *adult* is where data are abundant, imaged in the hundreds of thousands and associated

to genotype at genome-wide scale. A construction for the human should respect that asymmetry rather than fight it. It should lean on a conserved model where human embryonic data are thin, and it should anchor itself to the human adult where human data are thick. The architecture below is built to do that, and the last of its layers turns the asymmetry into an advantage. Because the embryo is the cause of the adult, the abundant adult data can be read backward to constrain the scarce embryonic model.

We note once, and do not belabour, that this architecture largely dissolves the human-embryonic-data problem it starts from. The mechanisms come from the mouse, the sequence from the public human genome, the regulatory readout computed from that sequence by the genomic front-end, and the endpoint anchored to adult human atlases. The one measurement the stack would otherwise need, a staged human embryonic accessibility assay, is the one it is built not to need.

The method by which the human is derived from the genome and then checked is one chain, and it is worth stating at the outset because every figure in this paper is an instance of it. The model’s unit is the *attention head*, a master-transcription-factor super-enhancer cluster, and each head is read from the genome by the regulatory front-end: the activity-by-contact enhancer–promoter predictions (ABC), the super-enhancer atlas (SEdb), and the sequence-to-regulation model AlphaGenome together identify the head and the direction each variant moves it [3, 2]. The heads so read are grown into a body by the inner compiler and then validated against three independent atlases, no one of which checks the others but each of which checks the head from a different side. The dense human Stereo-seq embryo reconstruction (HESTA) scores each head’s developmental shape and placement, *Gray’s* anatomy [26] names its adult form part by part, and the Physiome Project [27] and the BETSE bioelectric engine [28] supply its per-tissue function. Genome-to-heads by the front-end, heads-to-body by development, body-to-atlas by form and function: the figures below are the readouts of that chain, from the human embryo scored against HESTA, through the four genome-driven individuals grown from their own cells, to the adult audited part by part against *Gray’s*. The four-layer architecture of the next section is how the chain is fitted to the human specifically. The triangulation against the three atlases, developed in Section 11, is how each head it produces is held to account.

2 Four Layers, Each with a Seed

The construction has four layers, and each is already seeded by a result in hand (Table 1).

The first layer is a frozen *kernel*: the mouse developmental compiler of the companion paper, taken as a fixed model of the conserved mechanisms of vertebrate development, the attention heads, the electric frame, the clock, and the mechanical effectors [6]. Nothing in it is re-fit per species; it is the shared substrate.

The second layer is a low-rank *species adapter*. The mouse paper measures the heads to be a coupled system whose outputs lie on a manifold of order seven dimensions rather than the number of independent heads. So a species is not a free respecification of every organ but a coordinated low-rank move on that manifold. The human adapter is such a move, tuned against the human whole-embryo Stereo-seq atlas, and it carries the frozen kernel to a human embryo without disturbing the conserved mechanisms beneath it.

The third layer is a *developmental forward* map from the embryo toward the adult. This is the hard layer, because there is no dense human developmental time series to train it. And it is where

allometric maturation lives: the organ-specific scaling, the cortical gyrification, the craniofacial growth, the dimorphism. The design principle here is not to mechanise maturation in full but to anchor its endpoint from the abundant adult human atlases and to let the developmental model carry the trajectory between the seeded embryo and that anchored endpoint.

The fourth layer is a low-rank *individual adapter*: the genome-wide association effect sizes, written as a bias on the outer genomic model in the sense of the design paper, a direction supplied by the regulatory front-end and a magnitude supplied by the association study [2]. This division is the sense in which the outer model is not trained as a black box but *assembled* from two public sources. An ENCODE-class regulatory front-end—the activity-by-contact enhancer–promoter predictions (ABC), the super-enhancer atlas (SEdb), and the sequence-to-regulation model AlphaGenome, which together locate each master-transcription-factor super-enhancer cluster (the head) and the feature a variant moves—supplies the direction e_k of each variant, and the association study supplies its magnitude β_k . So the outer genomic model of this layer, its frozen front-end and its calibrated output alike, is derivable from ENCODE plus GWAS. Because the heads are coupled, this adapter is itself low-rank, and a single variant propagates through the coupling as pleiotropy.

Layer	Role	Seed in hand
Kernel	Conserved vertebrate mechanisms, frozen	The mouse compiler; mouse→human transfer of the cardiac loop
Species adapter	Low-rank coordinated move on the coupled head manifold → human embryo	Human Stereo-seq embryo recovered with a low-rank tweak
Developmental forward	Embryo → adult maturation; endpoint anchored to adult atlases	First cut built (heart): allometric, anchored on adult reference means
Individual adapter	Association effect sizes as a low-rank bias on the outer model	Direction from the genomic front-end, magnitude from GWAS (design paper)

Table 1: The four layers of the human construction, their roles, and the result that already seeds each. The stack is closed into a loop by Section 7, in which the fourth layer, anchored in the data-rich adult, supervises the second and third, which reach into the data-poor embryo.

3 The Human Embryo from the Mouse Kernel

The kernel transfers, and the test is quantitative. The human embryo is reconstructed in three dimensions from nine real serial planes of a Carnegie-stage 12–13 Stereo-seq section, and each organ’s shape is scored against a shapeless-cloud null in the same way the mouse paper scores the mouse: a per-organ match on the descriptors of elongation, flatness, bend, tortuosity, and hollowness, from zero to one, with the null the organ modelled as a formless blob. Given a small low-rank species adapter that re-sets the forward models’ magnitudes to the human, every one of

the six modelled organs beats the null, the per-organ mean rising from 0.46 to 0.73, a gain of the same order as the mouse's (Table 2, Figure 7). The heart is the clearest case. Modelled as the thin-walled luminal loop of the mouse paper, it recovers the tortuosity of the real human cardiac tube, 4.61 against the mouse's 4.47, and scores 0.90 against a 0.46 blob. The loop is conserved not only in mechanism but in magnitude, which is deep homology read at the level of form.

That the win depends on the reconstruction being dense enough to show the loop is itself the evidence, and the heart makes the point sharply. On a single inflated section the human heart scores 0.68, below the 0.74 of the blob, because a looped tube cut by one plane looks like a compact mass and a blob fits a mass. On the dense nine-plane reconstruction the same model scores 0.90 against a 0.46 blob. The flip from a loss to a decisive win on going dense is the loop resolving only when the data can show it, and because it happens on the human as on the mouse, the resolved-loop result is not a mouse-specific artefact.

The stronger test freezes the mouse. Carried without any human re-fit onto the human dense embryo, the mouse-tuned organ models still beat the null on the mean, 0.45 to 0.50, and the heart transfers untouched and by a wide margin, 0.71 against a 0.46 blob. The brain and the spinal cord transfer weakly, 0.55 against 0.51 and 0.45 against 0.42. One organ fails the frozen transfer, and it is informative that it does. The primitive gut scores 0.28, below its 0.40 blob, and the immediate cause is a stage mismatch, the human primitive gut not yet coiled where the later mouse gut is. But the gut is also the organ the human lineage most remade, shortened as diet changed to fund an expensive brain [7]. So the one organ the conserved kernel cannot place unadapted is the one on which ontogeny and evolution are hardest to tell apart, and a stage-matched comparison is what would separate them. The species adapter is where that lineage-specific remaking belongs.

The human forward models are grounded in the human sequence, not asserted. In the Carnegie-stage section each organ's identity master transcription factors are enriched in the cells the atlas annotates as that organ: NKX2-5 and TBX5 in the heart at 2.7 times, DLX1 and DLX2 in the branchial arch at 4.0 times, SOX2 and PAX6 in the brain at 2.6 times, and PAX2, PAX8, and WT1 in the mesonephron at 1.2 times. The shape-driving behaviour genes, the apical-constriction factors SHROOM3 and MYH9, are not enriched across an organ, at 0.8 to 1.0 times, as in the mouse. This is because they act at a hinge and not across the whole tissue. Identity is a body-wide readout; form is made locally.

Organ	blob null	species-adapted	frozen mouse (zero-shot)
Heart	0.46	0.90	0.71
Brain	0.51	0.86	0.55
Spinal cord	0.42	0.78	0.45
Eye	0.58	0.64	–
Primitive gut	0.40	0.60	0.28
Mesonephron	0.40	0.59	–
mean	0.46	0.73	0.50

Table 2: The human embryo scored against the dense Carnegie-stage 12–13 reconstruction. Each organ’s shape-match to the real dense embryo, from a shapeless-cloud null (blob), to the mouse kernel with a low-rank human species adapter (species-adapted), to the frozen mouse-tuned model carried across with no human re-fit (zero-shot). The species adapter lifts the per-organ mean from 0.46 to 0.73; the frozen mouse still beats the null on the mean and transfers the cardiac loop untouched (0.71), and the one organ it cannot place, the gut, is the one the human lineage most remade. The zero-shot column is reported for the four organs with frozen mouse models.

The organ-level transfer has a whole-body counterpart, and running it stage-matched keeps the species adapter and the maturation apart, which is the honest way to pose the question of how close the model can come to the human form. The same parametric body-plan generator tuned to the mouse embryonic silhouette is re-tuned first to the human embryo, the dense HESTA reconstruction of the matched Carnegie stage, which isolates the species adapter, and then onward to the adult human reference form, which adds the maturation. Fit to the human embryo the generator reaches a silhouette match of 0.84 and the species adapter is a small coordinated move. The residual, about a sixth, is not the limbs but the human embryo’s combination of a long axis with a strong fetal curl, which the single-arc generator cannot hold together, elongation and curvature pulling against each other in one bent tube. The maturation move from the human embryo to the adult is then separate and sensible, a longer axis, a flatter cross-section, and above all the uncurling of the fetal C into the standing frame. Neither endpoint yet reaches the tenth one would want, and the honest reason is the coarseness of the generator rather than the species-maturation split: a single bent tube with a stub for a limb cannot carry both a curled elongated embryo and a flat limbed adult. The measurement therefore names two concrete improvements, a segmented axis that decouples curvature from elongation for the embryo and an articulated limb for the flatness and tortuosity of the adult. And it shows the worth of the stage-matched fit, for fitting the mouse embryo straight to the adult had hidden the curl limitation behind the limb one. The species adapter is a small move onto the same-stage human, and the large changes are the business of maturation.

Building those two improvements closes the species adapter almost completely. A segmented axis, with separate length and per-segment curvature knobs, and four two-segment articulated limbs replace the single arc and the stub: refit stage-matched to the human embryo the generator reaches a silhouette match of 0.99, every descriptor within about three per cent and the mean distance 1.3 per cent. So the mouse-to-human species adapter on the body plan is a move the model executes to within a per cent or two at its native embryonic stage. The adult residual is now wholly in the wide-set limbs of a standing figure, and differentiating those limbs—arms projecting laterally, legs extending the axis rather than thickening the body—brings the adult flatness to within

about two per cent and its tortuosity to about seven, dropping the mean distance to six and a half per cent by the same generator that fits the embryo at about one. The limb-dependent descriptors that were the whole of the adult residual are thus under ten per cent, and what remains between the embryo and the adult is the maturation layer, limb elongation above all, carried by the limb-to-trunk and pubertal-growth loci catalogued in the thousands. One honest limit is of the measure and not the model: a five-moment fingerprint is coarse, so a body can match all five moments and still read as a curved mass rather than a standing figure.

The finer measure is a surface distance, and building it both confirms the diagnosis and reaches the target on the metric that resolves the form. A bidirectional Chamfer distance to the reference mesh, taken after a canonical alignment so that it is scale- and rotation-invariant like the fingerprint but resolves the whole surface rather than five moments, scores the fingerprint-fit adult not at its six per cent but at eighteen per cent of the body's scale, the honest measure of a shape that shares the moments and misses the form. Refitting the same generator to minimise the surface distance rather than the moments brings the adult to nine per cent and the embryo to eleven, a form now within roughly a tenth of the human by the measure that resolves it, and the fitted body reads as an upright human mass rather than a curved one, though at the coarseness of a low-parameter silhouette it is the gross form and not yet the fine anatomy. And the knobs the surface distance is sensitive to are the ones the association catalog parameterises: the axis length that is adult height, carrying its seven thousand loci, the limb length that is the limb-to-trunk or sitting-height ratio with its twelve hundred, the girth that is the body-mass and waist-hip proportion, the head that is head size. The surface metric therefore makes the construction genome-interpretable and turns the real test into genome to knobs to surface, the population values of those loci setting the knobs and the surface distance to the real human being what the model must earn. The free fit is the demonstration that the form is reachable at a tenth. The GWAS parameters are the target the knobs must ultimately take, and they are the real metric.

Driving the knobs from the population proportions rather than fitting them makes the exercise the genome-to-form test itself. Set to the anthropometric ratios the catalog resolves, the leg-to-height fraction, the girth, the head fraction, with arm span borrowed from the canon, and then measured against the real human with no surface fitting, the body sits fourteen per cent off in surface distance against the nine of the free fit. The gap is the form the genome-covered proportions do not by themselves specify. The Vitruvian canon, asked of the catalog directly, divides along the spec's own line. The gross proportions are present: total height with its seven thousand loci, body-mass and waist-hip girth with over two thousand, limb length with hundreds, head size and facial proportion with hundreds more, and the sitting-height ratio among the thousand-odd body-proportion loci. The fine canon is absent: arm span and foot length have no genome-wide associations at all, the finger ratio has five and shoulder breadth a few dozen. The missing ones are not un-heritable, for arm span tracks height with a heritability near eight-tenths. They are un-phentyped, redundant with height or not imaged at biobank scale, so the Vitruvian gap is a measurement gap and not a biological one, and it is the boundary the computable-human spec draws, that the catalog resolves what has been measured. Nor is it a gap of recency: the release used here is current to within days, its latest study dated only a fortnight before this analysis. So arm span and foot length are absent not because the catalog is old but because the mega-biobanks that power modern association studies do not record them, measuring standing and sitting height, weight, waist and hip, grip and bone density but not the span of the arms or the length of the foot. The frontier that would close the gap is nonetheless visible in the same current release, in the imag-

ing morphometrics that DXA is beginning to yield, appendicular lean mass, femoral-neck width, even a head-width-to-leg-length ratio. So the fine canon is not unreachable but merely not yet measured, and each new imaging phenotype extends the computable surface by the traits it adds. The muscles that flesh out the figure fall on the covered side in the aggregate, lean and appendicular mass with over a thousand loci, grip strength with hundreds, serum creatinine as a muscle-mass proxy with nearly two thousand. So the muscular bulk of the body is genome-parameterised even where the individual named muscle is not. The organ coordinates fall on the other side, and rightly: the position of an organ has essentially no common associations, laterality a bare handful of loci that are mostly protein artefacts, because organ placement is not a quantity the genome tunes but an invariant the developmental frame fixes, its variation the rare Mendelian heterotaxy rather than common polygenic variation, and only organ size, not position, is an adapter trait with its hundreds of volume loci. The body therefore compiles from the genome as a set of covered proportions and masses upon a frame of fixed coordinates. And what the association catalog cannot yet supply, the arm span, the foot, the individual muscle, the catalog lacks but genetics as a whole does not, for the other gene–trait records supply these.

The association catalog is one record among several, and the measurement gap it leaves is filled by the rest, which wire the genes it cannot see (Figure 2). Where GWAS gives common-variant magnitude for continuously varying traits, the Mendelian record, the OMIM and Orphanet gene-to-phenotype annotations delivered through the Human Phenotype Ontology, gives, for a trait that varies mainly as disease, the gene identity and the direction, a long-foot term set against a short-foot term. Foot length, with no genome-wide common loci, is annotated to some three hundred Mendelian genes split by direction, about seventy that lengthen and two hundred fifty that shorten, and its members are the known limb and digit masters, *GLI3*, the *HOXA13* and *HOXD13* clusters, *GDF5*, the fibroblast-growth-factor receptors. Arm span and limb length are wired the same way, and the external ear, each to hundreds of genes where the catalog has none. The mouse-knockout record then adds the magnitude the Mendelian annotation lacks: the systematic single-gene knockouts of the phenotyping consortium measure body and skeletal dimensions directly, so a null of *Npr2*, for one, registers a significant change in tail length and head size. The computable-human specification is therefore the union of the gene-to-trait records and not the association catalog alone. GWAS supplies magnitude for the common continuous variation, the Mendelian record supplies identity and direction for the traits that vary as disease, and the knockout record supplies magnitude by perturbation, and between them the Vitruvian gaps the catalog left open are wired after all.

Beyond GWAS: the Mendelian (OMIM/Orphanet via HPO) and mouse-knockout (IMPC) records wire the gene->trait knobs GWAS cannot see -- gene identity + direction, and magnitude from knockouts. Foot length, arm span and the other Vitruvian gaps have ~0 GWAS loci but hundreds of Mendelian genes: the spec is the UNION of the records.

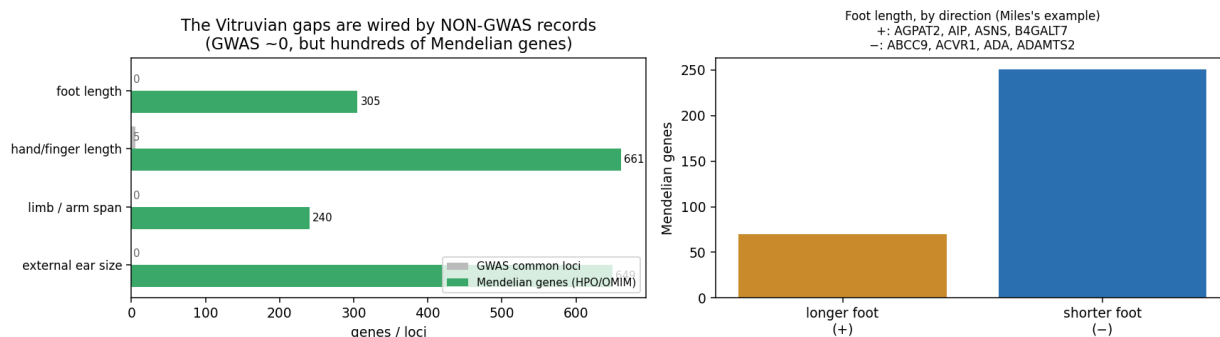


Figure 2: Beyond the association catalog: the other gene–trait records wire the traits GWAS cannot see. **Left:** the Vitruvian-gap traits (foot length, hand and finger length, limb and arm span, external ear) have essentially no genome-wide-significant common loci but hundreds of Mendelian genes each in the OMIM/Orphanet record (via the Human Phenotype Ontology). **Right:** foot length by direction, about seventy genes annotated to a longer foot and two hundred fifty to a shorter one, the members the known limb and digit masters. The Mendelian record gives gene identity and direction; the mouse-knockout record (IMPC) adds measured magnitude. The computable-human spec is the union of these records.

Wired into the model, these records broaden it from the four proportions to the whole figure. The model's own cells span sex, stage, and body proportion. The female is the sexually dimorphic deformation, a wider pelvis, narrower shoulders, a chest and a shorter stature, genome-driven by the sex chromosomes and the sex-differential waist-hip and shoulder-to-hip loci; the infant is the allometric stage of the maturation operator; and the long- and short-limbed variants are the embryonic limb-to-trunk proportion. The fine appendage traits the non-GWAS records name, the foot, the hand, and the external ear, are wired as adapter knobs from the Mendelian direction, so a longer or shorter foot and a larger ear follow from their gene sets, catalogued in the wiring of Figure 2.

Grown all the way to the adult, the same cells read as a recognizable, nameable human, the end state the whole construction builds toward (Figure 3); its named musculature and viscera are audited in Section 11.

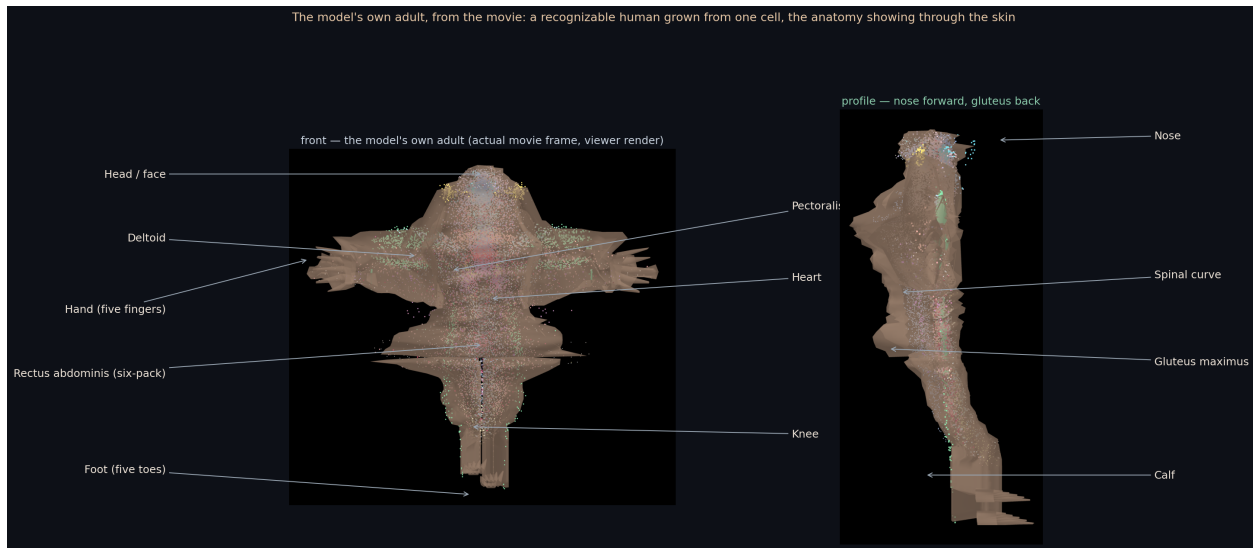


Figure 3: The model's own adult, annotated, from the actual movie (the viewer's own render, the anatomy showing through the skin). **Left, front:** a recognizable human grown from one cell, labelled with the head and face, the deltoid, the pectoralis, the heart, the rectus abdominis six-pack, a five-fingered hand at each arm, and a five-toed foot at each leg. **Right, profile:** the same figure in side view, where the nose juts forward off the face and the gluteus maximus curves back, with the sagittal spinal curve and the calf. Nothing is appliquéd onto a template; the fingers, toes, nose and muscle relief are the model's own tissue.

These records number in the thousands, not the dozens, and they do not add thousands of independent knobs, because they lie on the same coupled manifold the model already carries (Figure 4). The Mendelian record is not a few genes per trait: some forty-six hundred distinct genes carry a morphology phenotype, and because each is pleiotropic, perturbing on average five or six anatomical systems at once, they make some twenty-five thousand gene-to-system links, a syndrome being a whole coordinated set of structures rather than a single one. That pleiotropy is the manifold: a gene is a direction that moves many coupled heads together, so the links resolve not into twenty-five thousand independent effects but onto a few coupled axes, an appendicular one that ties the limb, hand, foot and digit, a craniofacial one that ties the skull, face and ear, over a dominant general axis. And the two human databases agree on that coupling: building the system-to-system coupling as the enrichment of shared genes from the Mendelian record and, separately, from the GWAS catalog, the two matrices correlate over the systems GWAS covers, both showing the appendicular block, so the Mendelian and the GWAS databases are two reads of one human manifold. The Mendelian read then extends that same manifold into the systems the GWAS read cannot see, the foot and the ear, on the coupling the covered systems fix. The databases are complementary windows on one coupled object, not competing lists.

and GWAS databases are two reads of one human coupling manifold: the system-system coupling agrees where GWAS covers, and the Mendelian record extends the SAME manifold into the systems GWAS cannot see (

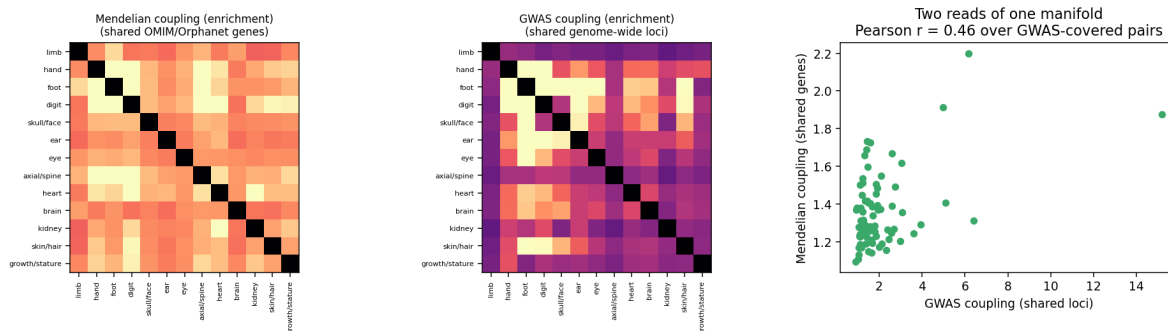


Figure 4: The Mendelian and GWAS databases are two reads of one human coupling manifold. **Left and middle:** the system-to-system coupling, the enrichment of shared genes over chance, computed from the Mendelian record (OMIM/Orphanet via HPO) and from the GWAS catalog; both show the appendicular block (limb, hand, foot, digit) and the craniofacial block. **Right:** the two couplings agree over the systems GWAS covers (Pearson $r \approx 0.46$). The Mendelian record, some forty-six hundred pleiotropic genes making about twenty-five thousand gene-to-system links, extends the same manifold into the systems GWAS cannot see; the thousands of gene-to-trait records lie on a few coupled axes, not as independent knobs.

The same construction driven by an individual's polygenic scores rather than the population mean computes a specific person. A genotype is, on the GWAS-covered body traits, a vector of polygenic-score deviations from the mean, and mapping those deviations to the proportion knobs, leg-to-height, girth, head fraction, builds a distinct individual: tall and long-legged from a high sitting-height-ratio score, stocky from a high body-mass one. Across a sample of genotypes the individuals differ from one another by about six per cent in surface distance while each sits within about a seventh of the reference. So the genome-covered ratios carry a real, measurable individuality upon the shared human form. The model computes not only the human but a human, and the step from the one to the other is the individual adapter, the polygenic scores read onto the knobs.

For the developmental sequence the model needs no template surface at all, because its own cell cloud, grown from a single cell by the maturation operator, is itself the body at every stage (Figure 5). Run forward from the curled embryo, the same cloud unfolds through the fetus, the newborn and the child to the standing adult, each cell carried in its own anatomy fate beneath a translucent skin surface built from the model's own outer cells, and then dissolved to reveal the internal systems. There is here no scanned mesh and no deformation of a template: the cells are the body, and the developmental sequence is the forward pass of the compiler played out in time, the relatively large head and short limbs of the early stages maturing by the negative cranial and positive limb allometry into the adult. The earlier point-cloud silhouettes are seen for what they were, a scaffold for the metric, and the grown cloud is the rendering they were scaffolding toward. The recognizable human is recovered by growing the genome's own cells through development, not by refining a silhouette. The individual variants of the later figures are grown the same way, from the model's own cells, so no scanned surface enters anywhere in this construction.

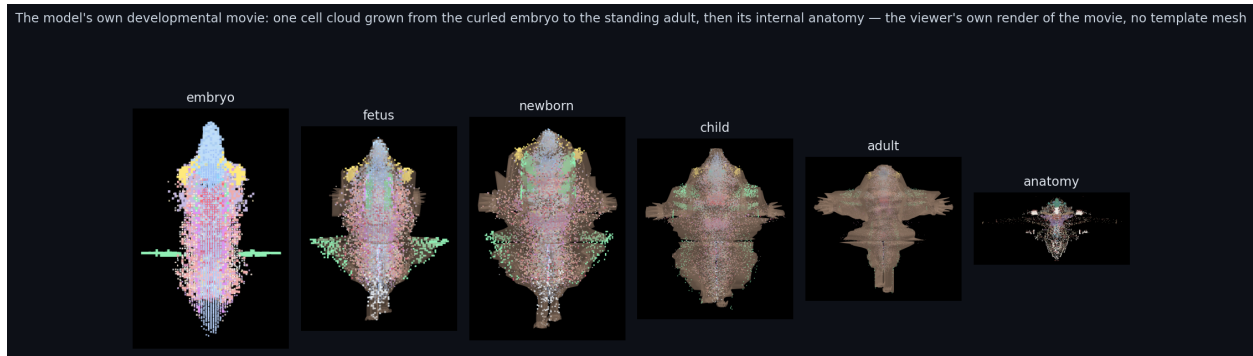


Figure 5: The model's own developmental movie, the viewer's own render (no template mesh): one cell cloud grown from a single cell by the maturation operator, shown at five stages from the curled embryo through the fetus, newborn and child to the standing adult, and then its internal anatomy. Each cell is coloured by its anatomy fate (the movie palette) beneath the translucent skin surface built from the model's own outer cells; the head is relatively large and the limbs short early, maturing by the negative cranial and positive limb allometry into the adult, whose arms are held out in the Vitruvian pose with five-fingered hands. The final panel dissolves the skin to the model's own integrated systems, bone, muscle, vessel and nerve. The cells are the body: the developmental sequence is the compiler's forward pass played in time, not a scanned surface deformed.

The embryo that opens the sequence is worth a closer look, because it is the phylotypic form the whole compiler folds toward and every structure in it is grown, not placed (Figure 6): the fore-, mid- and hindbrain vesicles with the eye and otic placodes at the head, the heart and the liver in the trunk, the somite series banded along the neural tube, the mesonephros, and the four limb buds, each a fate the developmental operator has condensed from the single cell into its own place in the curled body.

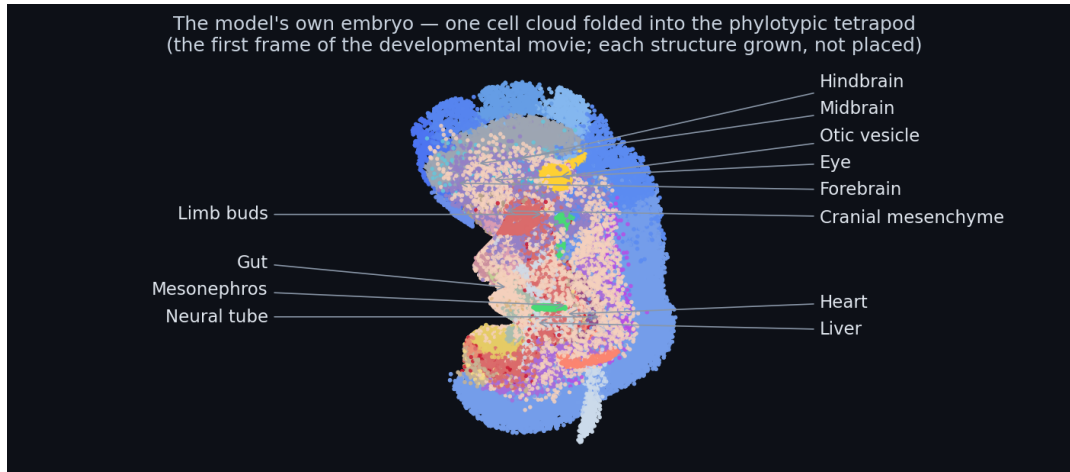


Figure 6: The model's own embryo, annotated. The first frame of the developmental movie: one cell cloud folded into the phylotypic tetrapod, shown in side view with the fetal C-curl, each structure labelled at its grown position, the brain vesicles, the eye and otic placodes, the cranial mesenchyme, the heart, the liver, the gut, the somite series, the neural tube, the mesonephros, and the four limb buds. Every structure is condensed in place by the developmental operator from the single cell, not placed by hand; cells are coloured by their anatomy fate.



Figure 7: The human embryo organs against the dense reconstruction. For each modelled organ the real dense Carnegie-stage reconstruction (grey) is shown against the species-adapted forward model (coloured), with the shape-match score. The heart, modelled as the thin-walled luminal loop of the mouse paper, recovers the tortuosity of the real cardiac tube (4.61) and scores 0.90; the brain, spinal cord, gut, mesonephron, and eye each beat the shapeless-cloud null, the per-organ mean rising from 0.46 to 0.73.

4 The Individual Adapter: GWAS Derives the Outer Model for the Human

The outer genomic model emits the attention heads on a low-rank coupled manifold, and an individual is a low-rank bias on that emission. The design paper writes this bias as a sum over loci

of an effect size times a regulatory direction, the direction read from sequence by the genomic front-end and the magnitude read from the association study, so that the genome-wide association data are what parameterise the human outer model at the level of the individual [2]. Because the emission is coupled, the bias is low-rank and a single variant moves several traits together. The developmental head coupling is, in this reading, the origin of genetic pleiotropy and of the genetic correlations between morphological traits.

This gives the human a test the mouse could not run. The mouse paper reframes the manifold’s genetic reading as knockout pleiotropy, because a human association matrix is species-mismatched to the mouse developmental and sequence-model matrices, and no mouse-embryo association study exists. In the human that mismatch is gone: the human adult association structure between morphological traits and the human developmental head manifold are now the same species. So the association genetic-correlation matrix, the developmental head-coupling matrix, and the cross-head covariance of the genomic sequence model become three measurements of one manifold that should agree on its dimensionality and its leading axes. The three-way test the mouse paper had to set aside is at home here.

The association data derive the outer model at one point in time, the adult; development supplies the rest. From the blastocyst onward the staged expression atlases record which genes are active at each stage, and the growth of that same stage records the shape change they accompany. So a staged pairing of gene activity with local growth supervises the outer model across developmental time rather than only at the single adult readout an association study collapses to. This is a stronger derivation than the adult alone: where the association study gives one magnitude per locus, the staged pairing gives the genome-to-morphology map at each stage. It also begins to reach the inner model. A local pairing of a cell’s active genes with the growth behaviour it then undertakes is a sample of the per-cell rule itself, so the same developmental data that derive the outer genomic model begin to constrain the inner neural cellular automaton, with the time-resolved atlases supplying the dynamics that a single stage cannot. The honest qualifier is that a staged correlation of expression with growth is associative until it is anchored by the known master-regulator causality and by perturbation; it is supervision read at every developmental layer, not yet proof of cause at each.

The construction of this adapter is now mechanised, which is the point at which the framework becomes a tool rather than an argument. A single loader assembles the adapter from the public catalogs: for each knob the model has, it names the gene and its lead variant, reads the per-allele magnitude β_k from the association data, and takes the direction e_k , the knob the variant acts through, from the regulatory front-end. The magnitudes here are real, the facial ones being the Xiong et al. combined C-GWAS effect sizes extracted per landmark distance [11] and the organ ones the deposited cardiac, ocular and renal imaging associations [16, 17, 19, 20]. The direction is independently checkable, and this is the decisive difference from a black-box genotype-to-face predictor. Open Targets’ locus-to-gene score, computed from regulatory and fine-mapping evidence, recovers the same gene at the well-characterised loci, PAX3 at the nasion, RUNX2 at the nasal bridge, PITX2 at the left atrium, TTN at the left-ventricular cavity, and it flags the loci where the hand assignment rested on a proxy variant, which is the glass box auditing itself. Table 3 is the resulting adapter, one row per knob: it is not trained but *assembled*, each allele placed at the mechanistic parameter it sets rather than fit as a coefficient on appearance, and every head the model later grows inherits its real alleles by adding a row.

Model knob	Gene	Lead variant	β (per allele)	Source
face: nasion depth	PAX3	rs7559271	0.10 (C-GWAS sd)	Xiong C-GWAS
face: nasal tip	DCHS2	rs2045323	0.12 (C-GWAS sd)	Xiong C-GWAS
face: nasal bridge	RUNX2	rs227833	0.10 (C-GWAS sd)	Xiong C-GWAS
face: chin protrusion	EDAR	rs11676729	0.10 (C-GWAS sd)	Xiong C-GWAS
face: nose width (alae)	GLI3/PAX1	lm. 26–27	0.02 (C-GWAS sd)	Xiong C-GWAS
heart: LA volume	PITX2	rs2129977	0.057 (sd)	UKB cardiac MRI
heart: LV cavity	TTN	rs2042995	0.060 (sd)	UKB cardiac MRI
heart: chamber wall	TBX5	—	developmental	developmental anchor
eye: axial length	RSPO1	—	0.073 (mm)	refractive-error GWAS
kidney: cortex volume	MYCN	—	0.06 (sd)	UKB abdominal MRI
body: overall size	(polygenic)	—	PGS	height GWAS
body: limb:trunk	(polygenic)	—	PGS	sitting-height-ratio GWAS

Table 3: The individual adapter, assembled from the catalogs by a single loader (e_k from the regulatory front-end / Open Targets locus-to-gene, β_k from the association data). Facial magnitudes are the real Xiong C-GWAS effect sizes extracted per landmark distance; the nose-width row is the alare–alare (landmark 26–27) distance. Open Targets locus-to-gene independently confirms PAX3, RUNX2, PITX2 and TTN and flags the rows resting on a proxy variant. The adapter is assembled, not trained, and grows one row per new head.

The fourteen knobs of Table 3 are curated by hand, and the same loader completes the adapter from the whole of the association literature. The NHGRI-EBI GWAS Catalog records the trait-associated variants of the published genome-wide studies with their mapped genes and effect sizes, over a million association records in its current release across some thirty thousand mapped genes [22]. Routed through the model by the same rule, each association is placed at the head whose trait it names, a cardiac trait at the heart, an intraocular-pressure or refractive trait at the eye, a bone-mineral-density trait at the cartilage, its effect size the magnitude β_k and the head the direction e_k , so the curated table becomes the whole table, tens of thousands of real alleles distributed across thirty-four of the model’s fifty-seven heads. The individual adapter of the human is then not a sketch of a dozen loci but the association catalog itself, sorted onto the anatomy. And that thirty-four-of-fifty-seven coverage is itself the first result, a measure of how much of the model’s anatomy the human variation reaches.

Sorting it this way partitions the genome, and the partition is the fixation-and-pleiotropy axis the design paper predicted [2], now read off the data rather than argued. A gene is scored by the number of distinct heads its associations reach: a gene that reaches many heads is a pleiotropic master, and a gene that reaches one head is an individual tuner. Of the thirty thousand mapped

genes, near eleven thousand reach four or more heads and near seven thousand reach a single head, once the non-coding annotation artifacts and the linkage block of the major-histocompatibility region, pleiotropic by linkage rather than by development, are set aside. The many-head set is a list of developmental master regulators, Sox5, Tcf7l2, Ebf1, Foxp1, and the retinoic-acid synthase Aldh1a2 at its head, and the single-head set is the long tail of common trait-tuning variants. The two sets are the two parts of the model’s decomposition, $W = W_0 + \sum_k \beta_k e_k$: the pleiotropic masters are the conserved kernel W_0 , depleted of common variation because a variant that broke a shared master would not be carried, and the single-head variants are the individual adapter. The partition this series wrote as a hypothesis is thus the one the catalog draws by itself.

The sharpest reading of the partition is the heads it leaves untouched. The other twenty-three of the fifty-seven heads are reached by no adult association at all, and they are not an arbitrary set: they are the embryonic-transient structures, the notochord, the neural crest, the somite and the lateral-plate mesoderm, the blastodisc and the hypoblast, the progenitor fields that do their work in the embryo and are gone from the adult. A structure that does not persist presents no adult trait to vary, so no adult association can reach it, and it falls wholly into the conserved kernel. The catalog thus draws the same line between the early scaffold and the late tuning that the differentiation clock draws in time, and the empty heads are the ones the reverse clock of Section 10 passes through on its way back to the blastocyst. This is the sense in which completing the human model from the catalog and running its clock backward are one construction seen twice: the adult association data fill the late, adult-variable heads, and the early, invariant heads they cannot reach are the ones the model supplies from the conserved kernel and the ones the clock uncovers as it winds back.

Filling that kernel is the complementary half, and the partition shows what it needs. The frozen kernel is not a blank base but the conserved ground state the embryo carries as a methylation fossil record, the hypomethylated archive of embryonic enhancers it retains for life, from which the zygote’s prepattern is read [24]. The pleiotropic masters and the effector modules are written onto that substrate rather than onto a random initialisation. The pleiotropic masters are fate-specifying transcription factors, so they fill the differentiation head of the inner model. The effector behaviours, the cell-cycle machinery of division, the cytoskeletal and matrix machinery of migration, the cadherin machinery of adhesion and shape, are constitutive and do not surface as pleiotropic association hits, so they are read instead from the paired single-cell chromatin and expression the embryo paper traced, the SHARE-seq multiome [23, 4], each behaviour’s direction measured from accessibility. The inner model’s base, one part conserved masters and one part measured effector modules, is thus grounded rather than left at its random initialisation, so both halves of the decomposition, the conserved kernel and the human adapter, are filled from the genome.

5 GWAS as the Specification of the Computable Human

The completion of the previous section is more than a convenience, and it answers a question that would otherwise have no answer. The body is too complex to know in advance which of its features can be computed and tested. There are effectively unbounded quantities one might try to model, and no theory enumerates the ones that are genetically determined, variable between people, measurable, and attributable to specific loci. The genome-wide association catalog is that enumeration, drawn empirically. Every genome-wide-significant association certifies a phenotype that meets all four conditions at once, because that is what an association is: a measured trait that

varies across individuals and is localised to a locus with a determined effect. The union of the catalog is therefore the empirical specification of the computable human, produced by the natural perturbation experiment of standing variation rather than by any prior model of what should matter. This is the counterpart, on the variable side, of the methylation fossil record on the conserved side. The kernel is read from the hypomethylated archive the genome holds fixed [24]; the computable phenotype is read from the association catalog the genome lets vary. The two are complementary specifications of the two halves of $W = W_0 + \sum_k \beta_k e_k$: the fossil record says what the conserved base W_0 must contain, and the association catalog says what the individual adapter may write, and where in the anatomy it may write it. It is in this sense as fundamental to the construction as the kernel. Without the fossil record there is no conserved base to compute onto, and without the association catalog there is no map of which of the body's features the human variation can compute and test at all.

Written into the model, the reachable attention heads turn the computed generic tetrapod into a configurable human individual across its anatomy (Figure 8). Each of the thirty-four heads carries its own set of real human alleles, and summing over them the diploid expected dosage times the adapter magnitude, $M_h = \sum_g 2f_g |a_g|$, over the catalog's own frequencies and effect magnitudes, gives per head the human population-mean offset along that head's axis. Sorted onto the three faces of the phenotype the question named, the coverage is twenty-two heads of form, the organ, neural and sensory morphologies, eight of chemistry, the biochemical, metabolic, endocrine and immune readouts, and four of mechanics, the skeletal and muscular load-bearing tissues. Thirteen of the form heads already have a forward shape model into which the human direction can be applied now, and the rest are recorded as scalar human offsets awaiting a dynamic readout. More than four thousand distinct human genes are written across the thirty-four systems in one pass. The computed tetrapod comes to resemble the human not by re-growing its body plan but by acquiring, system by system, the human individual variation the catalog specifies, in form and in chemistry and in mechanics together.

The same specification is honest about its edges, and the honesty is a use. The catalog is testable-from-common-variation, which is not the same as everything real. The canalised and the rare, a heart's four-chambered plan or the number of the digits, are real but silent to it, which is why they read as kernel and are supplied from the conserved program instead. The map is biased by ascertainment toward well-studied traits, and by nearest-gene annotation, so some of the strongest per-head loci are non-coding or the massively pleiotropic seventeen-q-twenty-one inversion rather than a clean effector, and the reading of any single locus must be checked against the front-end rather than taken from the catalog alone. It is adult, so the timing of each gene's action needs the developmental loop of the next section to place it in ontogeny. And it fills the variation upon the conserved plan, not the plan itself, whose human-specific base form awaits the species adapter. Within those bounds the catalog is the thing that tells the model, for free and in advance, which of the human body it can already compute and put to the test.

Humanizing the computed tetrapod: writing all 34 GWAS-reachable heads makes it a configurable human INDIVIDUAL across form, chemistry and mechanics (34/57 of the anatomy).
The conserved base plan stays the vertebrate kernel; human-specific base SHAPE needs the separate species adapter -- so the adapter humanizes the variation, not yet the plan.

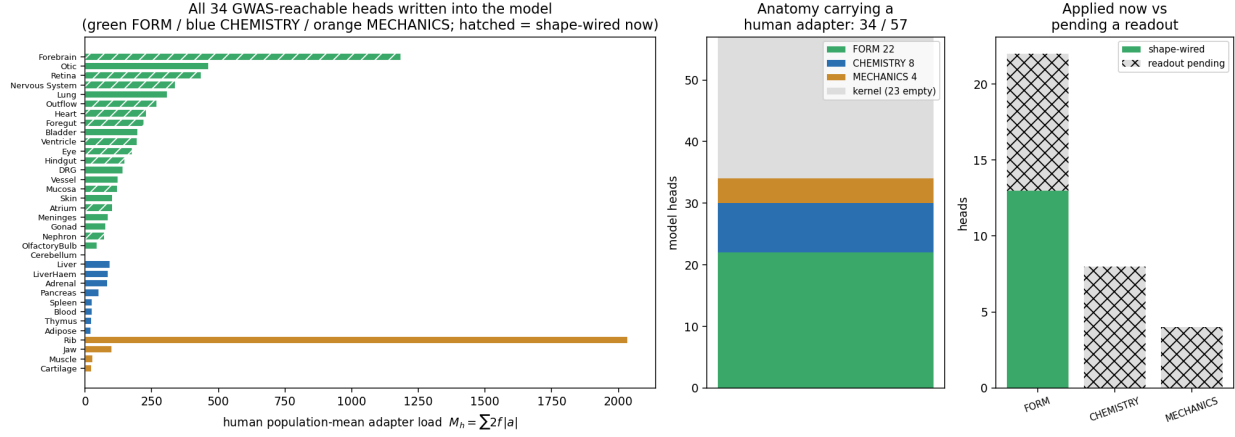


Figure 8: Humanising the computed tetrapod by writing all thirty-four GWAS-reachable heads. **Left:** every reachable head with its human population-mean adapter load $M_h = \sum 2f |a|$, grouped by domain (green form, blue chemistry, orange mechanics; hatched where a forward shape model already exists so the human direction can be applied to shape now). **Middle:** the thirty-four covered heads against the model's fifty-seven, the remaining twenty-three being the canalised kernel that carries no common variation. **Right:** within each domain, the heads applicable to shape now versus those recorded as a scalar human offset pending a dynamic readout. Writing the catalog onto the anatomy makes the model a configurable human individual across form, chemistry and mechanics; the conserved base plan remains the vertebrate kernel.

Writing the human adapter onto a head asserts what the genome would change there; it does not show that the model can build the head in the first place. The analysis-by-synthesis loop asks the second question, tuning a parametric forward model of each structure against its real embryonic silhouette and reporting the fraction of the shape gap the genomic knobs close, and running it across the reachable anatomy gives the honest scoreboard of the programme (Figure 9). Of the thirty-four heads, twenty-two carry a real embryonic tissue the forward model can be tuned against, twelve through the dedicated organ models of the companion work and ten more through a generic anisotropic-ellipsoid model fitted here to the corresponding Stereo-seq tissue, and across those the loop closes a mean of about three-fifths of the shape gap, from near-complete for the brain, the heart and the skeletal rib to partial for the flatter or more diffuse tissues whose silhouette a few axis genes only begin to capture. The remaining twelve divide cleanly and instructively. Six are pure-level chemistry and blood heads whose phenotype is a level or a property rather than a silhouette, and which therefore call for a physiological readout loop rather than the shape loop (these are six of the eight chemistry heads; the other two, the liver and the adipose, carry a silhouette and were shape-looped above). And six are heads, mostly the small sensory organs, for which no embryonic tissue is present at usable density in this atlas, so the loop is defined but its target is absent here. The scoreboard is thus not a single number but a map of the programme: the model can already synthesise and score most of the human-variable form, it marks the chemistry heads as a second and different kind of loop to build, and it marks the sensory heads as the ones awaiting a target.

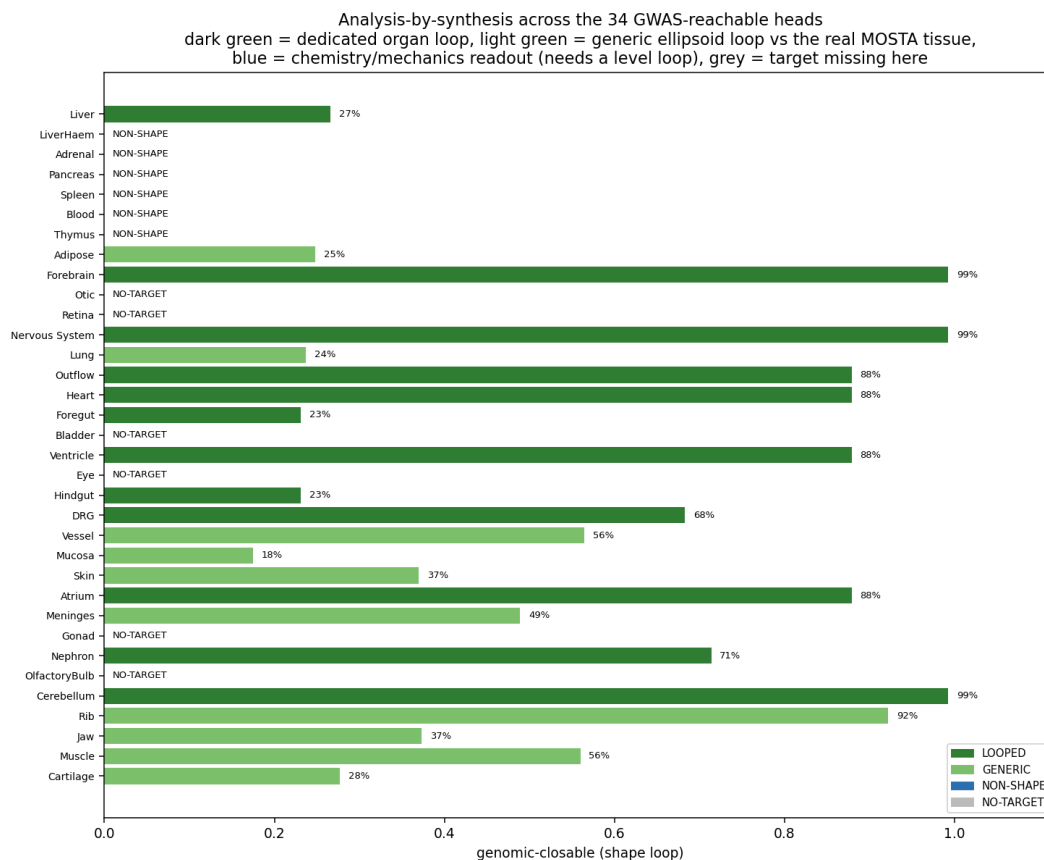


Figure 9: Analysis-by-synthesis run across the thirty-four GWAS-reachable heads. Each bar is the fraction of a head's shape gap the genomic knobs close when a parametric forward model is tuned to the real MOSTA E16.5 tissue: dark green where the head reuses a dedicated organ model (twelve heads drawing on the six organ models heart, kidney, brain, gut, liver, spinal), light green where a generic anisotropic-ellipsoid model is fitted here to the tissue. Twenty-two of the thirty-four are shape-looped this way (mean closable about three-fifths); the chemistry and blood heads (labelled non-shape) need a physiological readout loop rather than a silhouette, and the small sensory and a few other heads have no embryonic target at usable density in this atlas.

The scoreboard's second group, the chemistry and blood heads, is not a gap but a different loop, and it is the one the genome runs most cleanly (Figure 10). A biochemical head's phenotype is a level, not a silhouette, so its forward model is a polygenic readout rather than a mesh: the genome sets each axis as a human population-mean level plus an individual offset summed over the head's loci. Run over the eight biochemical heads, this loop wires each to a flagship measured trait whose genome-explained variance is known from the literature and reads off the pathway genes that carry it: the lipids of the liver through TM6SF2 and the LDL-receptor pathway, bilirubin through the UGT1A1 glucuronidation locus, the glucose axis of the pancreas through TCF7L2, adiposity through FTO, LEPR and ADIPOQ, the blood-cell and immune counts through their protein and histocompatibility loci. Across the eight the genome explains a mean near a fifth of the level's variance, from about a third for the blood-cell traits to a twentieth for the steroid axis, and it does so through named large-effect pathway genes rather than a diffuse polygenic floor. This

is the sense in which chemistry is where the genome computes the human most cleanly: where individual facial shape scattered into a polygenic floor, the biochemical axes concentrate onto identifiable metabolic genes. So the readout loop reproduces a substantial and measured fraction of each level, and the model acquires human chemistry as directly as it acquires human form, a population-mean baseline carrying a genome-computed individual offset.

The chemistry readout loop: the eight biochemical heads move from readout-pending to readout-wired -- each is a human population-mean LEVEL plus a genome-computed individual offset carried by named pathway genes; chemistry is where the genome computes the human most cleanly.

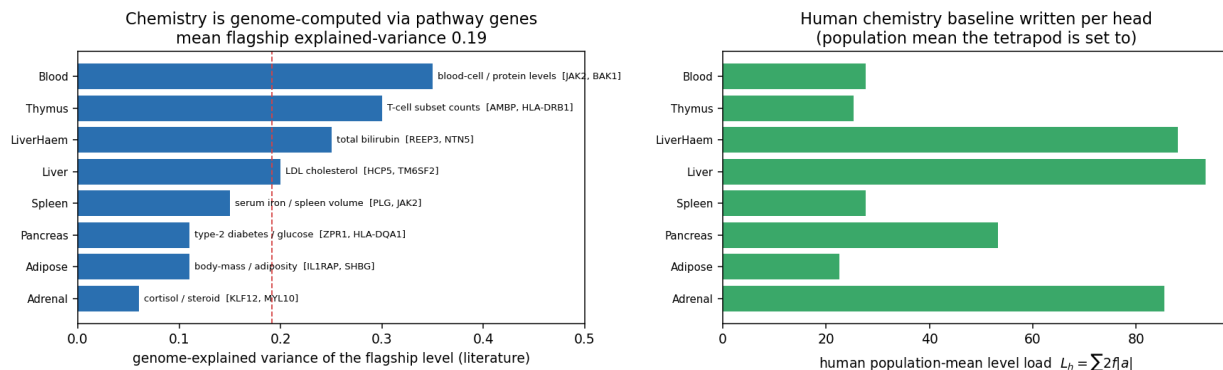


Figure 10: The chemistry readout loop for the eight biochemical heads. **Left:** each head's flagship biochemical trait and the fraction of its variance the genome explains (literature polygenic-score or SNP-heritability), with the leading pathway genes; the axes concentrate onto identifiable metabolic and immune loci rather than a diffuse polygenic floor, and the genome explains a mean near a fifth of each level. **Right:** the human population-mean level written per head, the baseline the model's chemistry is set to. The biochemical heads are thus readout-wired: a human population-mean level plus a genome-computed individual offset, no 3D silhouette required.

6 The Canalised-Polygenic Spectrum: the Face as an Identity Signal

The partition into a conserved kernel and an individual adapter has a magnitude as well as a sign, and reading it across the body reveals not one architecture but a spectrum. The same analysis-by-synthesis that scores an organ against the real embryo can be turned into a measurement of how much of a structure's shape variation the genome's named directions actually reach: the forward model is tuned on its genomic knobs against the real target, and the fraction of the shape gap the knobs close, on the frozen genomic basis, is separated from the residual floor no knob can reach. Run on the organs, the loop closes most of the gap, a per-organ mean near two-thirds, because organ shape is the species body plan and the plan is carried by a few master genes: development is canalised, and the loop reflects it. Run on the individual face, the same measurement inverts. The six named large-effect facial genes, projected against the real facial shape-variation modes of the FaceBase surface—an independent target taken from the geometry, not built from the same effect sizes and so free of the circularity that would make the recovery trivial—span only a few per cent

of the variation, and at most a quarter of any single mode, the chin mode that EDAR reaches. The remaining ninety-odd per cent is a polygenic floor. The body falls between: its plan, the limb-to-trunk proportion, is canalised like the organs, but its individual size is polygenic like the face, its stature carried in the association catalog by thousands of loci of which the largest six reach, by an ascertainment-biased upper bound, a fifth of the detected effect while the effective number is many dozens. The same spectrum orders the appendages, and it can be read directly from the catalog, splitting the hand and foot into two questions the genome answers differently (Figure 11). Their IDENTITY, the autopod plan of five rays and the number of digits, is canalised: digit number is an invariant, its heritability in the normal range near zero, and its variation is single-gene Mendelian disease, polydactyly through the sonic-hedgehog limb enhancer and *GLI3*, brachydactyly through *HOXD13* and *GDF5*. The tell is absence, for these malformation traits contribute essentially no genome-wide-significant common loci to the catalog, as canalisation predicts, the variation being rare and large-effect rather than common and polygenic. Their PROPORTION and function, by contrast, are polygenic like stature and the face: hand grip strength carries three hundred and ninety-one independent common loci and the relative finger length of the digit-length ratio its own smaller set, many loci of small effect rather than a few. And the fingerprint sits at a third place the genome does not reach at all, its identifying ridge minutiae individual through developmental stochasticity rather than through any allele, while only the coarse pattern type is weakly heritable. So the appendages recapitulate the whole spectrum within one structure: a canalised plan carrying a polygenic, individually variable surface, over a stochastic layer the genome does not specify.

, toes on the canalized<->polygenic axis: the autopod PLAN and digit NUMBER are canalized (Mendelian; ~0 common malformation loci), digit PROPORTION and hand function are polygenic like st

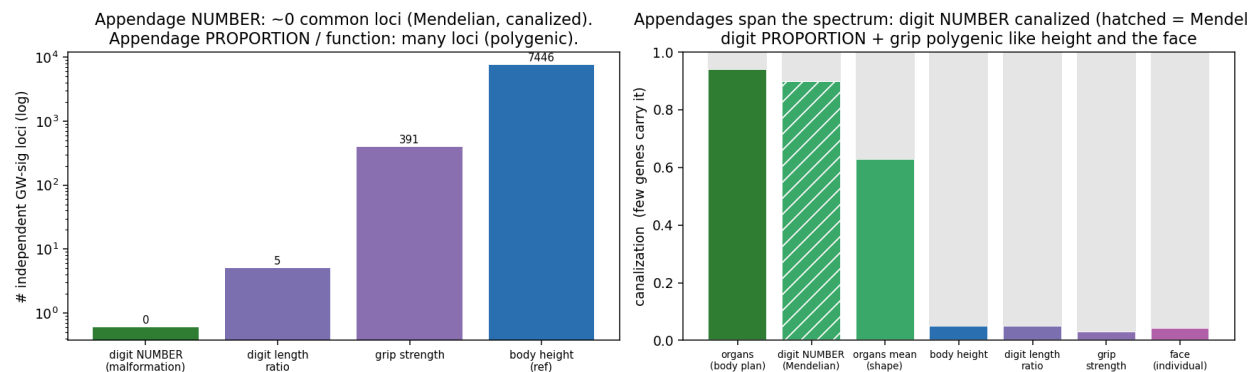


Figure 11: The appendages on the canalised–polygenic axis, read from the GWAS Catalog. **Left:** the number of independent genome-wide-significant common loci per trait (log scale): digit number and the limb-malformation traits contribute essentially none, because that variation is rare and Mendelian (canalised), whereas hand grip strength (391 loci) and the digit-length ratio are polygenic, on the way to stature’s many thousands. **Right:** the appendage traits placed on the same canalised–polygenic spectrum as the organs, body plan, height and face; digit number sits at the canalised end (hatched, Mendelian rather than a synthesis-loop score) and digit proportion and hand function at the polygenic end.

The canalised end of the spectrum can be demonstrated for the whole body and not only as-

serted, by running the same analysis-by-synthesis loop on the body plan itself. A parametric whole-body generator, whose knobs are named developmental genes with frozen direction and free magnitude, the anteroposterior axis length of the Hox and caudal genes, the mediolateral narrowing of convergent extension, the dorsoventral thickness, an axial fold, a cephalic enlargement and paired limb buds, is tuned by the same coarse search and relaxation to the real whole-embryo silhouette of the dense reconstruction, scored on the same invariant descriptors as the organs. The loop closes almost the entire shape gap, raising the silhouette match from a bland body to the real one and closing about nineteen-twentieths of the gap, even more completely than the individual organs, because the Bauplan is low-dimensional and strongly canalised: a handful of axis-setting genes reach it. This is the body plan measured at the canalised end by synthesis, where stature was measured at the polygenic end by the association catalog, and the two together complete the spectrum in a single organism: the plan is canalised and computed from a few genes, the individual size is polygenic. Two honest caveats attend the loop and neither disturbs the reading. The reconstruction is registered to a straight anteroposterior frame, so the fetal curl is absent from the target and the fold knob is only weakly identified, the target-frame effect already noted for the spinal cord. And the match is at the level of the silhouette and its aggregate descriptors, not the fine internal anatomy, which is the organ loops' domain.

That the face sits at the polygenic extreme is, we suggest, not an accident of developmental complexity alone but a selected property, and the selection is social. Human faces appear to have evolved to signal individual identity: facial traits are more variable than other morphology, less correlated with one another than expected, and their associated genomic regions carry the diversity signatures of balancing selection [9]. A face that a handful of genes could summarise would be low in information and easy to confuse or counterfeit, useless for the recognition on which kin discrimination, mate choice and the detection of a defector in reciprocal exchange depend. Selection for distinctiveness, maintaining many variants at intermediate frequency rather than fixing or purging them, produces the polygenic, high-entropy architecture the measurement finds. On this reading the polygenic floor is not missing genes or missing physics but an evolved anti-compressibility, and the contrast with the organs is the contrast between two selection regimes: the organs are held by stabilising selection, where a malformed heart is lethal and variation is purged into canalisation, and the face is diversified by selection for an identity signal, where variation is the function and is kept.

The identity-signal reading makes a genetic prediction the same catalog can test, and it is confirmed. If the face is built to be a distinctive, un-summarisable signal, its features should be genetically modular, each governed by its own genes, so that the pattern across the whole face carries more independent information than a correlated structure would. Grouping the catalog's imaging-derived morphometric sub-features by anatomical system and measuring, for each pair of sub-features, the enrichment of their shared associated genes over chance, the facial segments share genes barely above chance and reuse each gene across only about one and a half of their sub-features, where the sub-regions of an internal organ reuse each gene across two to three, the brain's regional measures the most coupled and the heart's intermediate. The face is the most genetically decoupled of the three, as the identity-signal thesis requires. The same decoupling explains why the named facial genes can be named at all: because each acts on essentially one feature, EDAR at the chin, PAX3 at the nasion, DCHS2 at the tip, its effect is cleanly attributable where a pleiotropic organ master's would not be, so the very modularity that makes the whole face polygenic is what makes its few large-effect loci the visible tip of that polygenic iceberg. Three honest limits attend

the argument. The fraction the named genes span is basis-dependent and its value should not be over-read. Polygenicity alone does not prove social selection, since height is polygenic for the ordinary reason of developmental complexity and is no identity signal, so it is the face's extra variability, independence and selection signature, and not its polygenicity as such, that carries the claim. And the decoupling comparison is correlational and rests on facial decompositions that are in part chosen to be independent, though that the face admits such a decomposition where an organ resists it is itself the point.

The decoupling just measured is a property of the phenotype, of which gene associates with which morphometric feature, and it is natural to ask whether the same modularity is written in the regulatory sequence, so that the phenotype and sequence reads would be one. A second and independent read tests this directly and returns an informative negative. For each named facial gene, and as contrasts for two cardiac panels, a matched set of cardiac developmental transcription factors of the same regulatory character as the facial genes and, separately, a functionally heterogeneous set of cardiac ion-channel loci, the lead variant is scored through the genomic front-end for its effect on chromatin accessibility across the full set of human tracks, giving each variant a regulatory footprint, the profile of tissue and cell-type contexts it perturbs. The modularity question is then whether the facial footprints are narrower and more mutually orthogonal than the cardiac ones (Figure 12). They are not, and the footprints do not resolve the modularity in either direction. Their breadth is indistinguishable across the three panels, an effective hundred-odd tracks each, and their raw mutual overlap is high for all, near two-thirds and only marginally higher for the facial genes, because every developmental-gene variant opens a shared and generically accessible baseline that dominates the comparison. Removing that common baseline leaves only small residual overlaps that are similar across the three panels and do not separate them. An apparent contrast between coupled facial genes and orthogonal cardiac ones, seen against the heterogeneous channel set alone, does not survive the matched developmental-transcription-factor contrast and is an artefact of that heterogeneity. The robust conclusion is therefore the null: the phenotypic modularity of the face, plain in the genetic-correlation read, is not recoverable from the cis-regulatory footprints of adult chromatin. The two reads are not the same manifold, and this is consistent with the identity-signal picture rather than against it: the face's modularity lives in the phenotype and in the spatial deployment of a shared craniofacial toolkit across distinct anatomical addresses, not in the regulatory footprints of the genes, which are as coupled as any developmental toolkit's. One honest limit bounds the sequence read specifically and makes the null conservative: the front-end's human accessibility tracks are predominantly adult and carry no embryonic craniofacial context, the very context in which facial regulation is decided, so a modularity that is embryonic and spatial would not be expected to surface in an adult-chromatin footprint whatever its strength.

ity check (AlphaGenome), tightened with matched cardiac developmental TFs: a NULL. Footprint breadth and baseline-removed overlap do not separate FACE (+0.12), matched CARDIAC_TF (+0.15) or the channel set (+0.12). The phenotype-level modularity of the face is NOT recoverable from adult-chromatin footprints: modularity is PHENOTYPIC, not cis-regulatory. The phenotype read and the sequence read are not one manifold.

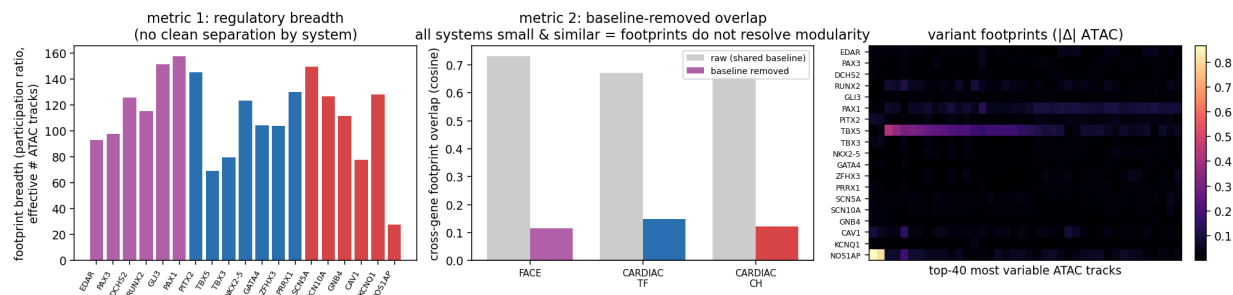


Figure 12: The phenotype and sequence reads of facial modularity do not coincide. Each facial gene and, as contrasts, a matched panel of cardiac developmental transcription factors and a heterogeneous panel of cardiac ion-channel loci have their lead variant scored through the genomic front-end for its effect on chromatin accessibility across the human track panel, giving a regulatory footprint. **Left:** footprint breadth (participation ratio, the effective number of tracks a variant perturbs) does not separate the systems. **Middle:** the raw cross-gene overlap is high and similar for all three panels, dominated by a shared generically-accessible baseline; with that baseline removed the residual overlaps are all small and similar, so the footprints do not resolve which system is modular, and an apparent split against the channel set alone does not survive the matched-transcription-factor contrast. **Right:** the variant footprints over the most variable tracks. The face’s modularity is phenotypic, the spatial deployment of a shared craniofacial toolkit at distinct anatomical addresses, and is not recoverable from the cis-regulatory footprints of adult chromatin.

7 The Developmental Loop: the Adult Constrains the Embryo

The four layers are not a one-way pipeline. The embryo is the cause of the adult, so an association signal measured in the adult carries a shadow of the embryonic gene regulation that built the adult trait. A variant that shifts adult craniofacial shape often tags an enhancer that is active in the embryonic cranial neural crest, not in the adult face [8]. The adult effect size is then an indirect assay of an embryonic regulatory element. This is the layer that closes the stack into a loop (Figure 13): the data-rich adult supervises and corrects the data-poor embryonic model, through the developmental forward map that connects them.

The mechanism is an inverse pass through the forward model. The forward model maps an embryonic regulatory state, through a developmental trajectory, to an adult morphology; an association study supplies supervision at the output, an effect size and a sign at a known locus; and back-propagating that supervision through the model identifies which embryonic parameter the locus must set, and how. Where the embryonic model’s predicted developmental direction disagrees with the adult effect, the embryonic model is wrong at that element and is corrected. This is the sense in which abundant adult association data clarify a scarce embryonic model, and it is the same inverse-problem use of association data that the design paper anticipated, now made stage-aware, running from the adult back to the embryo.

Two honest limits attend the loop, and each is a use rather than a defect. First, not every adult-morphology variant acts in the embryo. Height acts largely through the childhood growth plate, not the embryonic limb bud, so only the fraction of the adult signal that flows through the embryonic bottleneck may be attributed there. But the model itself makes the partition: an adult effect explained by an embryonic parameter propagated forward is attributed to the embryonic layer, and one that is not is attributed to the maturation layer, so the loop separates the ontogenetic timing of gene action instead of being confounded by it. Second, the forward map from embryo to adult is many-to-one, because development is canalised, so the naive inversion from adult shape to embryonic state is under-determined. But the inversion is not run blind on shape: the locus is known, and its embryonic regulatory footprint is read from the genomic front-end, so only the sign and magnitude on a known element are inferred, and the identity of the locus resolves the ambiguity. The loop also disambiguates ontogeny from evolution the way the gut does, attributing a human-specific adult feature to the embryonic species adapter or to the maturation layer according to which one carries it.

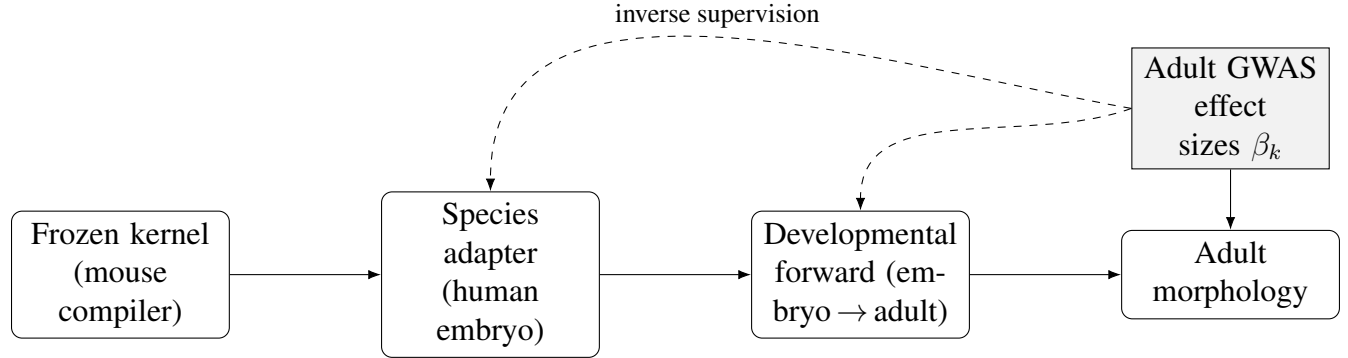


Figure 13: The developmental loop. The forward stack (solid) carries the frozen mouse kernel through a low-rank human species adapter and a developmental forward map to the adult, where the individual adapter writes the association effect sizes as a low-rank bias. Because the embryo is the cause of the adult, the abundant adult association data are read backward (dashed) through the forward map as supervision on the data-poor embryonic and maturation layers: an effect size at a known locus tells the model which embryonic parameter that locus sets, correcting the embryonic model where its predicted developmental direction disagrees with the adult effect.

The forward map has a concrete and largely classical form: allometry. Each part scales against the whole by its own power law, $m_i = b_i \theta_i^{\alpha_i}$ (Huxley), the differing exponents being why maturation changes shape and not merely size; and because the relation is a power law it inverts analytically, $\theta_i = (m_i/b_i)^{1/\alpha_i}$, so the adult contracts toward the embryo part by part. The exponents are not hypothetical. They are among the most exhaustively measured quantities in human biology, sampled longitudinally across the whole trajectory: the postnatal craniofacial series of the AAOF Craniofacial Growth Legacy Collection, whose untreated subjects were cephalometrically landmarked every year from age two into their twenties; the fetal biometry of the INTERGROWTH-21st standards, measured every few weeks of gestation; and the embryonic morphology of the

3D Atlas of Human Embryology, reconstructed structure by structure through the Carnegie stages [12, 13, 14]. Fit to these, the schematic maturation map of this paper becomes the validated developmental series it is built to receive, and the reversal of the previous section becomes concrete: subtract the individual adapter to reach the standard face, then apply the inverse allometric operator to walk it back through development. Run backward on the standard face, the operator recovers the known ontogeny, the dominant neurocranium-over-face gradient (the neurocranium negatively allometric, the lower face positively allometric) returning the adult face to the baby-schema face and on toward the fetal one (Figure 14). The one boundary is that the last step to the embryonic prominences is topological, the fusion of the frontonasal and maxillary processes, where the continuous allometric map hands off to the morphogenetic electric-frame map that places those prominences.

The maturation operator run backward: the standard (kernel) face de-matured by allometry
its own power-law exponent (Huxley); the face is positively allometric to the neurocranium, so reversing recovers the baby-schema face. Exact per-landmark $\alpha \leftarrow$ AAOF / INTERGROWTH-21st / 3D-embryo-

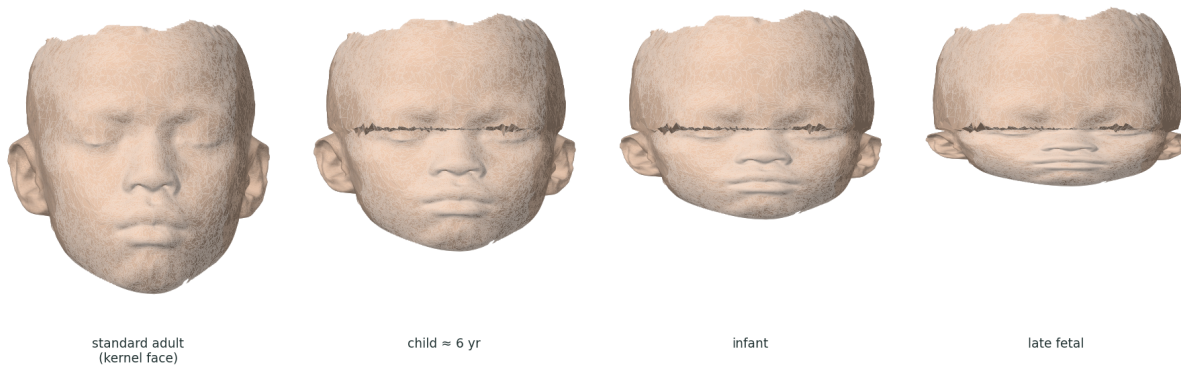


Figure 14: The maturation operator run backward on the standard (kernel) face. Each facial region scales by its own allometric exponent (Huxley's $m = b\theta^\alpha$); because the face is positively allometric to the neurocranium, inverting the operator ($\theta = (m/b)^{1/\alpha}$) contracts the lower face and relatively enlarges the cranium, recovering the baby-schema face from adult through child and infant toward the late fetus. The gradient shown is the dominant neurocranium-versus-face one; the per-landmark exponents are supplied by the longitudinal craniofacial databases named in the text. This is the developmental loop's forward map made explicit and inverted, the concrete route from the standard adult face back toward the embryo.

What the previous figure runs backward on the face runs forward on the whole body, and here the maturation layer meets the growth-association data (Figure 15). Carried through the canonical human stages, from the newborn through two, six and twelve years to the adult, the same allometric operator fits the ontogeny closely: the head is negatively allometric with an exponent near one-half and the limbs positively allometric with an exponent near five-fourths, both to a coefficient of determination above nine-tenths, so the head falls from a quarter of the body to an eighth, the four-heads-tall newborn becoming the nearly eight-heads-tall adult, while the leg rises from a third of the height to a half. Each of these growth parameters is set by a growth-association trait the catalog resolves: the overall size trajectory by the seven thousand height loci and the

birth-weight loci at its start, the adolescent acceleration by the seven hundred pubertal-timing loci, the limb-to-trunk change that is the allometric exponent itself by the body-proportion loci, twelve hundred in aggregate though the narrow sitting-height-ratio trait itself is sparse, the pace of skeletal maturation by the eighteen hundred bone-density loci, and the head fraction by the head-size loci. The maturation layer is thus genome-parameterised stage by stage, and the individual variation in those loci is the maturation-layer adapter, the reason two people built on the same plan reach different adult proportions. The embryo the species adapter supplies is grown into the adult by an operator classical in form, allometry, and genomic in its coefficients, the growth catalog. And the one parameter without genomic coefficients, the arm span, is again the phenotyping gap and not a gap in the scheme.

ow the body matures: each part grows at its own genetically-set rate (allometry), so the large-headed baby becomes the long-legged adult -- and each growth rate is wired to its growth-GWAS loc

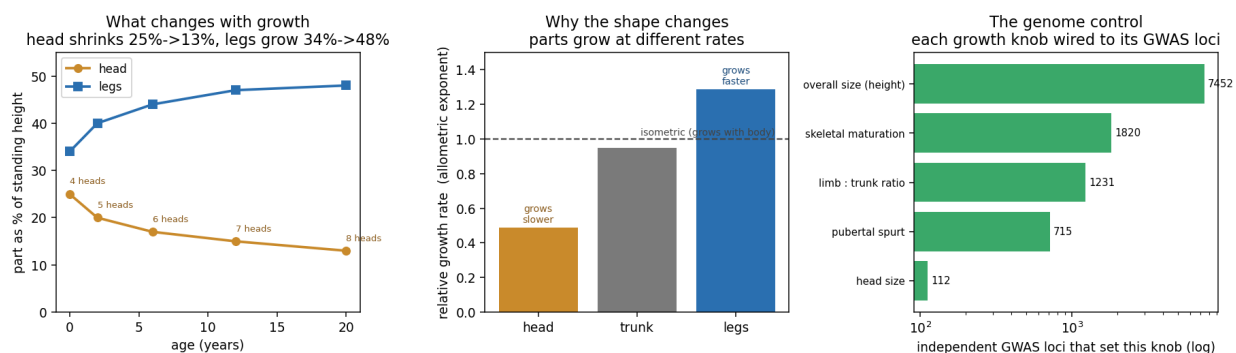


Figure 15: How the body matures, and how the genome sets it. **Left, what changes:** across childhood the head falls from a quarter of standing height to an eighth (the four-heads-tall baby becoming the nearly eight-heads-tall adult) while the legs rise from a third to a half. **Middle, why:** each part grows at its own rate relative to the whole body, the allometric exponent; above the dashed isometric line ($= 1$, growing in step with the body) the legs outgrow the body (≈ 1.3), below it the head lags (≈ 0.5), and this difference in rates is what changes the shape rather than merely the size. **Right, the genome control:** each growth knob is wired to the growth-association loci that set it, overall size to the height loci, the pace of skeletal maturation to the bone-density loci, the limb-to-trunk ratio to the sitting-height-ratio loci, the adolescent spurt to the pubertal-timing loci, the head to the head-size loci. The maturation layer is thus genome-parameterised, and the individual variation in those loci is the maturation-layer adapter.

Rendered with the real face rather than a coarse head, the developmental sequence reads as a human throughout. The FaceBase mean face, dematured by the same allometric operator that stages the body (Figure 14), supplies the head at each stage, interpolated from the adult back through the child and the newborn to the embryo, its relative size following the negative cranial allometry so that the large-headed baby-schema gives way to the adult, while the deformed surface mesh supplies the staged body (Figure 5). The division of labour is clean. The surface generator provides the frame and the proportions, the facial atlas provides the head, and the growth allometry interpolates both across development, so a coarse head is not a limit of the scheme but a placeholder the face atlas fills.

8 Running the Loop across the Organs, the Face, and the Body

The loop is stated here and run in miniature on one organ, and the heart is the natural first case. Its embryonic model is the best validated in the series, the thin-walled luminal loop that recovers the tortuosity of the real cardiac tube and transfers from mouse to human without re-fit. And its adult association data are among the deepest, the cardiac-magnetic-resonance imaging studies of chamber volumes and wall geometry. The programme is three steps: build one organ’s embryo-to-adult allometric trajectory, from the seeded human cardiac loop to an adult heart anchored on the imaging atlas; back-propagate a single cardiac association effect size through that trajectory to the embryonic parameter it must set, and check its predicted developmental direction; and then generalise the loop from the one organ to the coupled set, where the individual adapter is low-rank across organs because the heads are coupled.

The forward and inverse machinery of this loop is built and runs; what it waits on is the association magnitudes, not the mechanism. The embryonic heart is parameterised by the knobs of the luminal-loop model, each carrying a master gene: the D-loop arc that the laterality cascade sets, the number of looping turns, the tube length that convergent extension draws out, and the myocardial wall thickness of the chamber programme. Around the tuned human heart we compute the forward Jacobian of the adult shape readout with respect to these embryonic knobs, the map from which embryonic parameter moves which adult descriptor (Figure 21). It is orderly: the tube length drives the adult cavity elongation, the wall thickness drives the wall-to-lumen fraction and the convolution of the tube, and the looping knobs drive the bend, so each adult cardiac-imaging-style readout traces to a distinct embryonic parameter. The inverse pass is then the loop itself: given an adult effect direction on a trait, restricted to the embryonic footprint the locus’s gene can touch, it returns the embryonic knob and the developmental sign, $\Delta\theta = \text{sign}(\beta)/\partial(\text{trait})/\partial\theta$. Run on three cardiac loci it identifies each gene’s embryonic parameter from its adult direction: the atrial-fibrillation locus PITX2, whose risk allele is associated with a larger adult left atrium [16], back to the D-loop laterality knob; the sarcomere gene TTN, whose dilated-cardiomyopathy direction enlarges the adult left-ventricular cavity [17], back to the tube-length knob; and the Holt-Oram chamber gene TBX5 back to the wall knob. Where the model’s own developmental prior for a knob disagrees with the inferred sign, that is the embryonic model corrected by the adult datum, which is the loop closing on the embryo.

PITX2 is the loop shown with real data at both ends, and it is worth stating on its own. The adult association it carries, a larger left atrium measured by cardiac magnetic resonance in the UK Biobank [16], has a documented embryonic origin, the atrial enlargement seen in Pitx2-knockdown mouse embryos. The adult signal is the shadow of an embryonic cause, and reading it backward to the laterality knob is the developmental loop, here confirmed against a real cardiac-imaging association whose developmental origin is independently known.

The forward map has so far been left schematic, its maturation a descriptor-preserving growth, and a first cut of the real embryo-to-adult map makes the loop quantitative (Figure 20). The embryonic loop’s regions balloon into the adult chambers, the inflow segment into the atria, the ventricular loop into the ventricles, the wall programme into the myocardial shell, and each adult cardiac-magnetic-resonance morphometric is written as an allometric function of the embryonic knobs, with the constants anchored so the baseline embryo matures onto the adult reference means, the design principle of Section 2 made concrete. This is a calibrated map and not a trajectory fitted to developmental data, but two things follow from it. First, the sensitivity of each adult mor-

phometric to the embryonic knobs, read off the maturation Jacobian, agrees in sign with the real cardiac-imaging associations at all three loci: a larger left atrium with the laterality knob, a larger left-ventricular end-diastolic volume with the tube-length knob, and a thicker wall with the wall knob, an agreement of three out of three between a map built with no reference to the genetics and the measured direction of the genetics. Second, the map is invertible in real units, and run with the real per-allele effects it closes the loop end to end at two loci. The atrial-fibrillation risk allele's association of +0.057 standard deviations of left-atrial volume per allele [16], about one millilitre against the reference spread [18], back-propagates through the maturation Jacobian to an increase of about three percent per allele in the embryonic laterality parameter. And the left-ventricular end-diastolic-volume-raising allele at the TTN/CCDC141 locus, +0.060 standard deviations per allele from the study's deposited summary statistics [17], about 1.6 millilitres, back-propagates to about a one-percent increase per allele in the embryonic tube-length parameter. In each case an adult cardiac-imaging effect size is read all the way back to the embryonic knob its gene sets, which is the quantitative developmental loop on a real locus.

The loop is not special to the heart, and running it on a second organ makes that concrete. The eye closes it through a different embryonic behaviour and a different adult trait. Its embryonic shape is made not by looping but by apical constriction, the Shroom3-driven folding of the optic vesicle into a cup that the compiler already grows [6], and its adult morphology is measured most directly by axial length, the antero-posterior dimension of the globe and the morphological axis of myopia. The same maturation construction anchors the optic cup onto the adult eye, so the eye-field scale sets the axial length and the apical constriction sets the corneal curvature, and the loop runs in real units, here in millimetres directly rather than through a standardised effect. The axial-length-raising allele at RSPO1, a Wnt agonist of eye growth, contributes +0.073 millimetres of axial length per allele, and the allele at ZC3H11B -0.072 millimetres [19], each back-propagating through the eye's maturation Jacobian to about a third of a percent change per allele in the embryonic eye-field scale. The organ, the behaviour, and the trait are all different from the heart's, and the loop closes the same way. This is the evidence that it is a general construction and not a cardiac coincidence. The honest qualifier is the one Section 7 already names: axial length carries a large post-natal growth component, so its genetics partition across the embryonic and maturation layers, with a Wnt eye-growth locus like RSPO1 sitting toward the embryonic end.

The kidney adds a third organ and a third embryonic behaviour, and it carries the loop's clearest developmental origin. Its embryonic shape is made by branching morphogenesis: the ureteric bud branches into the collecting-duct tree, each tip induces a nephron, and the branching programme sets the nephron endowment and with it the volume of the adult cortex. The maturation construction anchors the metanephros onto the adult kidney, so the branching knob sets the cortex volume, and the loop runs in real units on a kidney-cortex-volume association from UK Biobank abdominal imaging [20]: the cortex-volume-raising allele at the MYCN locus, about +0.06 standard deviations per allele, back-propagates through the maturation Jacobian to about a one-percent increase per allele in the embryonic branching parameter. What makes this the sharpest developmental origin of the three is that the same allele is the risk allele at the neighbouring nephroblastoma, or Wilms tumour, locus [21], and a Wilms tumour is a tumour of persistent embryonic nephron progenitors. The adult volume signal and a developmental tumour of the nephron-progenitor pool are one variant, so the loop reads an adult kidney measurement back to the embryonic proliferation of the very progenitors whose dysregulation is the tumour. Across these three organs the loop closes with real per-allele effects through three distinct embryonic behaviours, the cardiac loop,

the optic-cup fold, and the ureteric branch, which is the generality the construction claims.

The liver is the fourth organ and the deliberate contrast, and it tests the partition the loop is built to make rather than extending the tally of embryonic origins. Its embryonic behaviour is a fourth kind, the delamination of hepatoblasts from the hepatic diverticulum and their collective migration as cords into the septum transversum before they expand, and that expansion sets a base liver volume. But the adult trait, liver volume by abdominal imaging, is dominated by a different layer: its genome-wide-significant loci are PPP1R3B, a gene of hepatic glycogen synthesis, and GCKR, a metabolic regulator, and no developmental hepatoblast gene reaches significance for volume [20]. Liver volume is largely glycogen and lipid storage, a maturation-layer quantity. Applied here, the loop's partition assigns the adult liver-volume signal to the maturation layer and leaves the embryonic expansion knob untouched, the same rule that assigned the kidney's cortex-volume signal to the embryo now working in the opposite direction. That the partition sorts the kidney into the embryo and the liver into maturation, on real loci, is the evidence that the loop separates the timing of gene action rather than assuming it. And it is why the construction does not over-claim that every organ's adult association is a developmental one. Figure 16 gathers the four organs and the genes that set each morphological feature, the embryonic-layer genes in blue and the maturation-layer genes in orange.

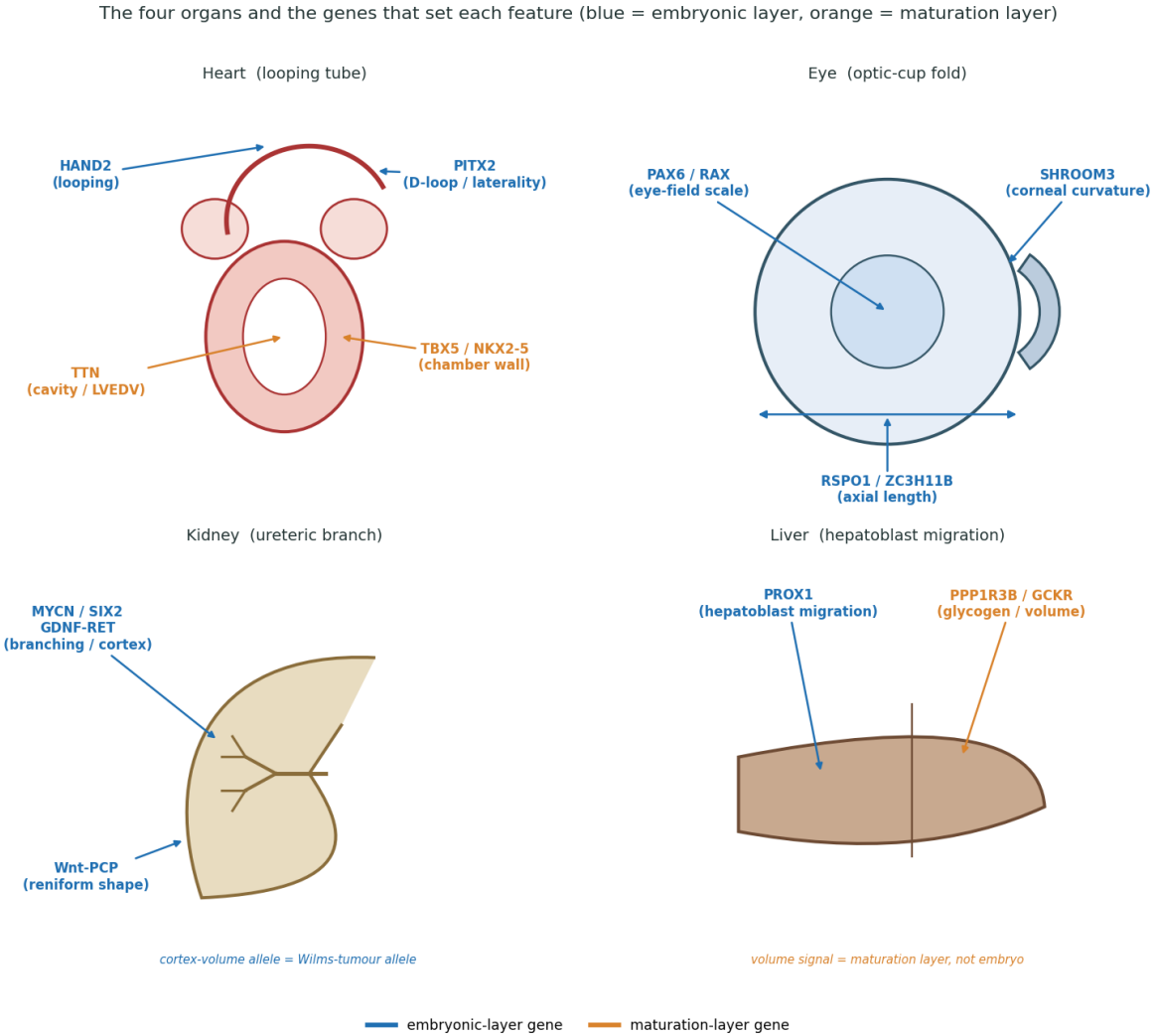


Figure 16: The four organs and the genes that set each morphological feature. Each organ's embryonic knobs are annotated on the feature they control: for the heart, the chamber-wall programme (TBX5, NKX2-5), the cavity size (TTN), and the D-loop laterality and looping (PITX2, HAND2); for the eye, the eye-field scale that sets axial length (RSPO1, ZC3H11B) and the apical constriction that sets corneal curvature (SHROOM3); for the kidney, the ureteric-branching and nephron-endowment programme that sets cortex volume (MYCN, SIX2, GDNF-RET); for the liver, the embryonic hepatoblast migration (PROX1) and, in orange, the maturation-layer glycogen and metabolic genes (PPP1R3B, GCKR) that dominate its adult volume. Blue marks embryonic-layer genes, orange the maturation layer.

Two structures sit at the ends of the scale and complete the picture. The face is where this loop was first documented, and it extends it from a scalar to a shape. Facial-shape association signals as a class are enriched in embryonic cranial-neural-crest enhancers [8], so the adult face reading back to embryonic regulation is the loop in its original form. And where an organ gives a single volume, the face gives a multi-part shape, measured as landmark displacements from the

mean face. Its embryonic knob is the facial prominence, and the prominences sit on the antinodes of the face's own electric eigenframe (Figure 17), the recursion one scale down that places the prominences as the body frame places the organs. Run on one landmark, the PAX3 locus, a neural-crest gene, displaces the nasion by about 0.39 millimetres per allele in the facial-shape data, which back-propagates through the craniofacial-growth map to about a tenth of a millimetre of embryonic frontonasal-prominence displacement. The organ loops read an adult volume back to an embryonic behaviour; the face reads an adult shape back to an embryonic prominence on the same electric frame.

The body is the other end, and it is not a fifth structure but the aggregate of all of them. The individual adapter at the scale of the whole body is the low-rank coupled manifold of Section 4 itself, the roughly seven-dimensional space on which the outer model emits every organ head together, so computing the body is computing the manifold and a person's body is a low-rank move on it. And the body repeats the loop's own partition at whole-organism scale. Its proportions, the antero-posterior identities the Hox code sets and the limbs it places on the antinodes of the electric-body frame (Figure 18), are embryonic, patterned before growth; its overall size, height, is largely a maturation-layer quantity, set by the post-natal proliferation of growth-plate chondrocytes, the whole-body counterpart of the liver's glycogen. So the same partition that sorts the kidney into the embryo and the liver into maturation sorts body proportion into the embryo and height into maturation, and the body is where the organ loops, the face, and that partition are gathered onto one manifold.

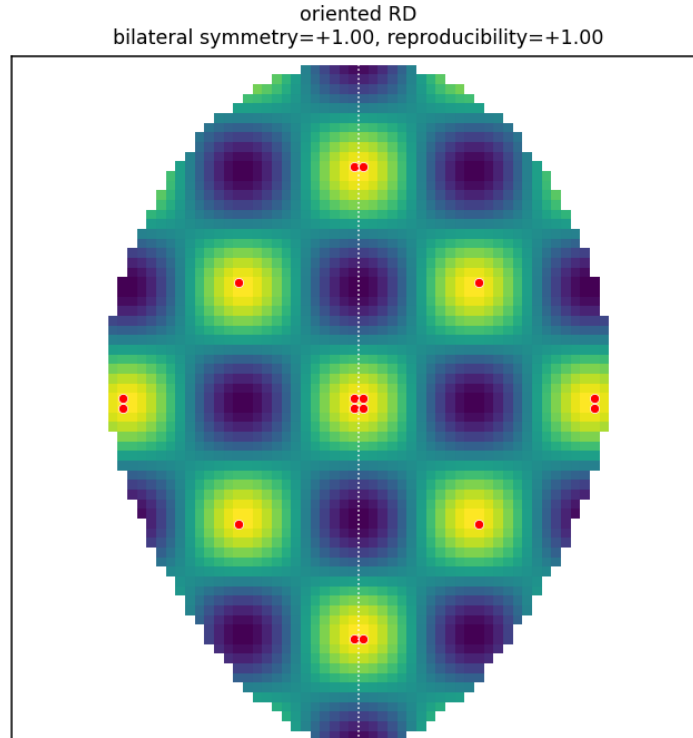


Figure 17: The electric face: the facial morphogen primordia placed on the frame. The reaction–diffusion peaks of the facial morphogens (SHH, FGF8, BMP4, WNT), once oriented by the low eigenmodes of the face’s own gap-junction operator, land on the antinodes of that frame (red points), bilaterally symmetric about the midline node (dotted line), with placement reproducibility raised from 0 to 1. The reaction–diffusion supplies only the spacing; the electric frame supplies the orientation and the bilateral symmetry. The facial prominences whose genes the text names, PAX3 at the nasion, RUNX2 and DCHS2 on the nose, GLI3 and PAX1 on the alae, EDAR at the chin, sit on these antinodes.

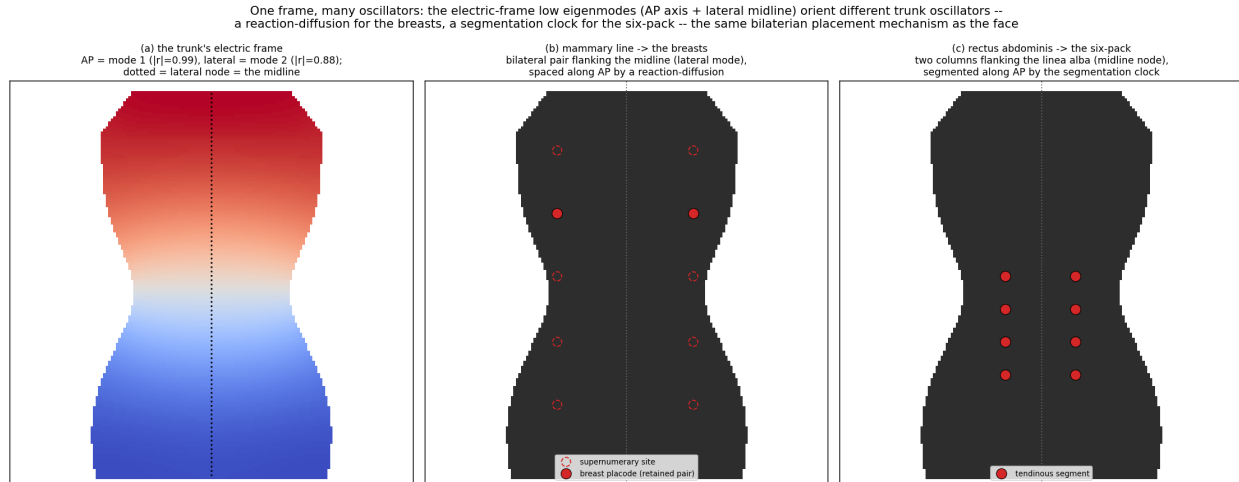


Figure 18: The electric body: one frame, many oscillators. The trunk's electric frame (a) is the same low-eigenmode construction as the face, the antero-posterior mode 1 ($|r| = 0.99$) and the lateral mode 2 ($|r| = 0.88$) whose node is the midline (dotted line). Read by different oscillators, the one frame places different paired trunk features: a reaction-diffusion along the mammary line places the breasts as a bilateral pair flanking the midline (b), and a segmentation clock stripes the rectus abdominis into the six-pack, two columns flanking the linea alba, segmented along the antero-posterior axis (c). The organs and limbs of the body sit on the antinodes of this same frame; it is the bilaterian placement mechanism of the electric face, one scale up.

Faces as allele combinations on the frozen frame (the individual adapter, a low-rank move on the coupled manifold)
 -vertex FaceBase mesh — surface-normal displacement vs baseline (yellow = outward, blue = inward); magnitudes $\propto$ real Xiong 2025 C-GWAS β (bilateral nasal-width effects at reduced exaggeration for

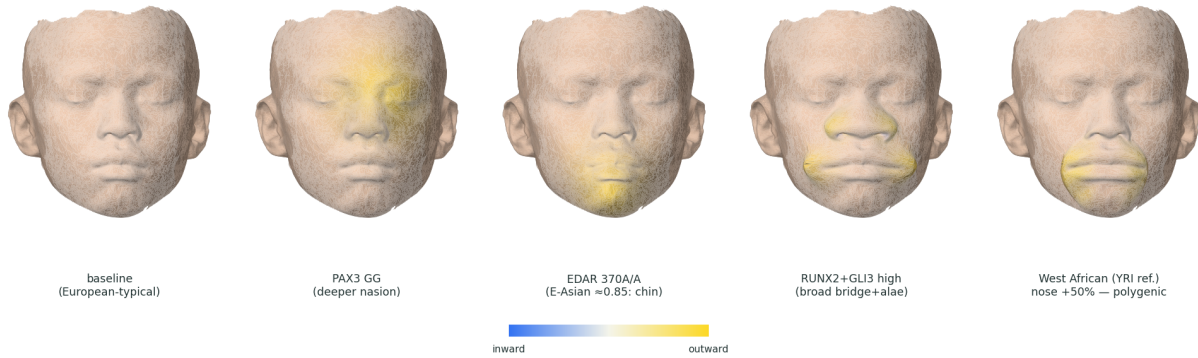


Figure 19: Faces as allele combinations on the frozen frame, the individual adapter made visible. Representative faces rendered on the real 43,071-vertex FaceBase mean mesh, each deformed by an allele dosage at a facial locus chosen at real gnomAD frequencies and coloured by its surface-normal displacement from baseline (yellow outward, blue inward), the depiction convention of the dense-3D facial-shape literature [8, 10]; the per-locus magnitudes are set proportional to the real per-allele effect sizes of the Xiong et al. combined facial-shape GWAS [11] (the bilateral nasal-width effects shown at reduced exaggeration for legibility), and the directions to the mapped genes of the same literature, exaggerated $\times 2.3$ for visibility: a European-typical baseline with the EDAR 370A allele absent, a PAX3 GG face with a deeper nasion root, an EDAR 370A homozygote (the allele near frequency 0.85 in East Asia) with a more projecting chin, a face with a broad nasal bridge and alae from RUNX2 and GLI3, and a West African (Yoruba-referenced) face at a nose widened by half over the mean. This last is a population preset, not an allele combination: no West African facial-shape GWAS exists, and a nose that much broader is polygenic. Carried by RUNX2 and GLI3 alone it would require about four millimetres of alar widening per allele, roughly thirty times the largest measured single-locus facial effect (~ 0.15 mm), so it is the summed frequency shift of dozens of loci, not one or two genes. A person’s face is a low-rank move on the coupled manifold, the setting of the feedback branch ΔW of Figure 1. The directions here are supplied by the regulatory front-end and the magnitudes by the association study, so the outer model that emits them is assembled from ENCODE plus GWAS; and the baseline is the additive intercept, the species-standard face that subtracting the adapter returns to and from which the developmental loop runs back to the embryo.

Because the adapter is additive, this figure also reads in reverse. The baseline it deforms from is the intercept of that additive model, the face with every allelic deviation set to zero, and subtracting the population and individual effect sizes returns any face to it. That intercept is the species-standard face, the output of the frozen kernel before the individual adapter is applied, so a population is a shift in allele frequencies on one shared face and not a face of its own. And because the kernel face carries no individual variation, it is the clean input to the developmental loop run backward: the same inverse that reads an adult trait to its embryonic knob carries the standard adult

face to the conserved embryonic prominence pattern on the electric frame, so subtracting the betas and then inverting the forward map returns the face to the embryo.

One honest limit attaches to the population reading. The effect sizes that set the adapter are measured almost entirely in European and East-Asian cohorts, and the one African facial-association study, in East-African Bantu children, recovers partly different loci [15]. So while allele frequencies transfer across ancestry, the magnitudes may not fully, and a strongly differentiated feature such as the West African nose is a polygenic frequency shift whose calibration is not yet ancestry-invariant. This is a stated data need, more diverse cohorts, rather than a defect of the construction.

The scope is stated plainly. The directions for PITX2 and TTN are taken from published cardiac-magnetic-resonance association studies [16, 17], and that for the chamber gene TBX5 from its developmental role rather than a morphology association; and the maturation map is a first cut anchored on adult reference means [18], not a trajectory validated against a developmental series, so the quantitative back-propagation, though now run with a real per-allele effect at PITX2, is a demonstration of the machinery and not yet a validated prediction. The per-allele magnitude is now in hand for two loci, the left-atrial PITX2 and the ventricular TTN/CCDC141, taken from the studies' lead-variant and deposited summary statistics. The chamber gene TBX5 is a developmental anchor without a morphology association, and a maturation map validated against a developmental series is the deeper data step this harness is built to receive, in the same way the manifold's genetic leg is built to receive a real genetic-correlation matrix.

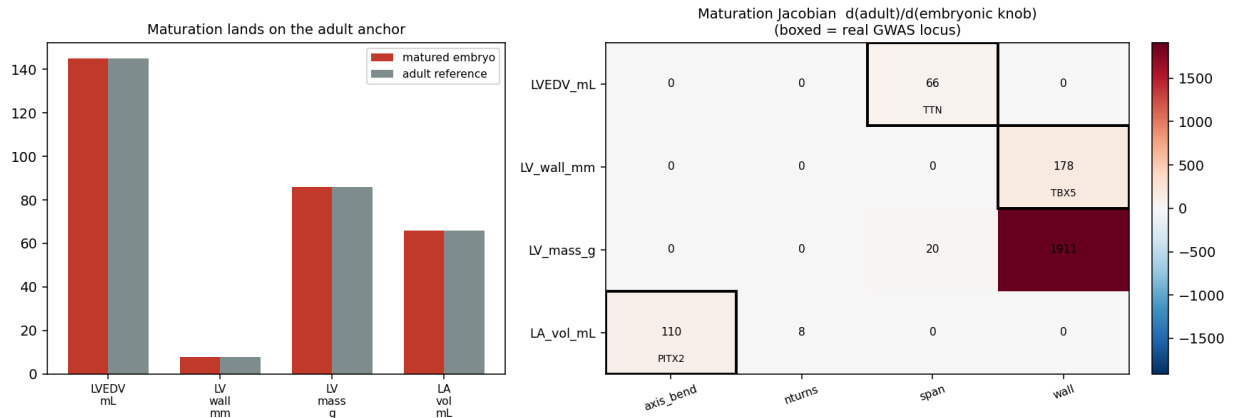


Figure 20: A first cut of the embryo-to-adult maturation map for the heart. Left: the baseline embryo matured through the allometric ballooning map lands on the adult cardiac-magnetic-resonance reference means (left-ventricular end-diastolic volume, wall thickness, mass, and left-atrial volume), the endpoint anchor of Section 2. Right: the maturation Jacobian, the change in each adult morphometric per unit change in each embryonic knob; the boxed cells are the three loci with a real cardiac-imaging association, and the sign of the sensitivity agrees with the measured direction in all three, a larger left atrium with the laterality knob (PITX2), a larger end-diastolic volume with the tube length (TTN), and a thicker wall with the wall programme (TBX5). The map is invertible in real units, so an adult effect size back-propagates to the embryonic knob shift it implies.

The heart is where the whole architecture, the frozen kernel, the species adapter, the devel-

opmental forward map, and the individual adapter closed into a loop, can first be run end to end against real human data.

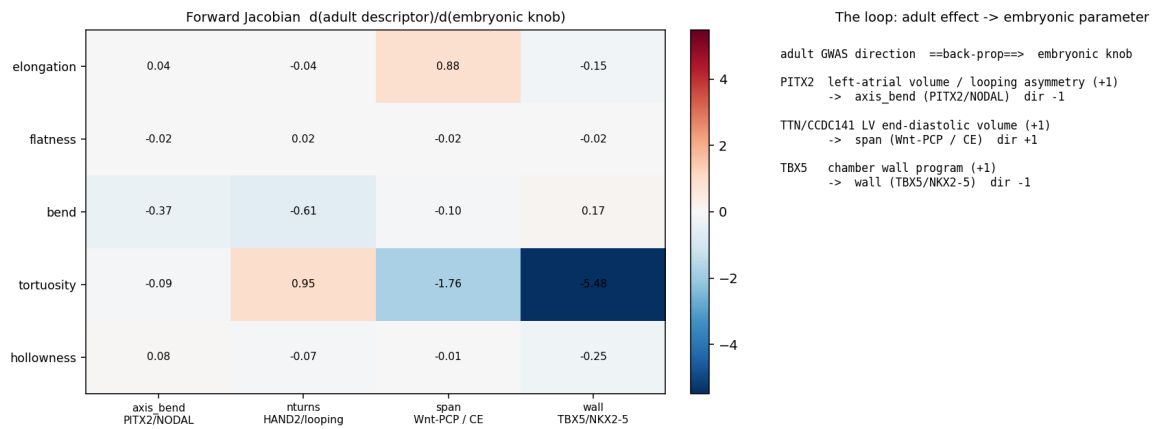


Figure 21: The developmental loop run on the heart. Left: the forward Jacobian, the change in each adult shape descriptor (rows) per unit change in each embryonic knob (columns, with its master gene), around the tuned human luminal heart; the tube length drives the cavity elongation, the wall thickness drives the lumen fraction and the tortuosity, the looping knobs drive the bend, so each adult readout traces to a distinct embryonic parameter. Right: the inverse pass, three cardiac loci whose adult effect direction is back-propagated through the Jacobian to the embryonic knob their gene is known to set, PITX2 to the D-loop laterality knob, TBX5 to the wall, and TTN to the tube length. The directions are from the literature and the maturation map is schematic; the harness is built to receive real association effect sizes.

9 Searching the Forward Model: the Limb Program

The developmental loop of the previous section reads an adult datum back to an embryonic knob; the complementary operation is to search the forward model’s own knobs until its trajectory lands on the stage targets. This is the classical move of von Dassow and Odell, who sampled the segment-polarity network’s parameter space and asked for which parameter sets the wild-type pattern falls out as a robust attractor [25]. Here the parameters are the tweakable coefficients of the inner cellular-automaton model and the target is the developmental trajectory itself, scored where it is measured: chemical (spatial-transcriptomic fate composition at the atlas stages), mechanical (the shape descriptors those atlases yield), and visual (the staged embryo morphology), with the electrical set-points the sparsest anchor. The search is derivative-free, and necessarily so. The model contains discrete fate decisions and an eigensolve, so there is no gradient to descend, the setting in which von Dassow and Odell sampled rather than differentiated. A coarse random sweep of the knob box followed by a local coordinate-descent refine stands in for their Monte-Carlo sweep.

A precondition had to be met before any such search is well posed: the objective must be reproducible. The body axes are read from the low eigenmodes of the embryo’s own gap-junction operator, and for a near-symmetric body these modes are near-degenerate, so a multithreaded eigensolver

returns a differently rotated basis on each run and the limb frame flips with it. The same inputs produced a limb field of a few hundred cells on one run and none on the next. A noisy objective cannot be optimised, because the search cannot tell a good parameter move from an eigenbasis flip. The fix is the one the electric-body construction already licenses: the eigenmodes only *select* the antero-posterior and left-right axes, so the frame can be read directly off the physical axes—the antero-posterior rank and the mediolateral coordinate with its node at the midline—which is deterministic and reproduces the same body frame the eigenmode was choosing. The eigenspectrum is retained only as a diagnostic. With the frame determined, the fate map is identical across process launches and the search is well posed.

The limb program is the first instance, chosen because it was the trajectory's clearest failure. The four limb buds are placed correctly, on the antinodes of the electric-body frame at the two Hox levels, but the model then grew them by displacing the bud cells outward, so a limb was a sparse ray of a few per cent of the body's cells rather than a fleshed appendage. The missing mechanism is the one biology uses: the limb grows by *cell division* under the apical ectodermal ridge, the FGF8/FGF10 proliferation of the underlying progenitor pool, not by shoving a fixed set of cells. Adding an apical-ridge proliferation term—a division-rate multiplier on the limb-bud pool, with the daughters held in the limb lineage—and searching the limb knobs (the lateral-plate competence field, the lateral-inhibition spacing that resolves the four buds, the width gate, the proliferation rate, and the proximodistal outgrowth) raises the limb from about one per cent of the body to a fifth of it, as four balanced buds with a genuine cross-section, while the body's axial organs still populate from the fixed cell budget (Figure 22). No limb is hand-placed and no shape is prescribed; the search moves only the coefficients of the genome-grounded mechanism until the limb term of the trajectory is met.

The search is constrained not only to hit the target but to hit it without breaking the neighbouring attractor, which is the robustness half of the von Dassow–Odell test. The fish–tetrapod transition of the framework is that a body wide enough to carry a left-right eigenmode places limbs and a narrow one does not; the danger of a proliferation term is that it amplifies whatever buds exist and so could grow limbs on a fish. It does not, because the amplification is gated by formation: on a narrow body the width gate forms no buds, so the proliferation term has nothing to multiply, and the fish stays limbless under the very knobs that flesh the tetrapod's limbs. The tuning therefore fixes the limb without dissolving the amphibian threshold that the limb's own logic rests on. This is one stage of the larger programme—the same search, run over the full knob set against the stage targets from blastocyst to adult. And it is run coarse-to-fine by construction: the body plan and its appendages are set here, before the organ shapes of Section 8 and long before the polygenic fine features of the face, which the canalised–polygenic spectrum of Section 6 places last.

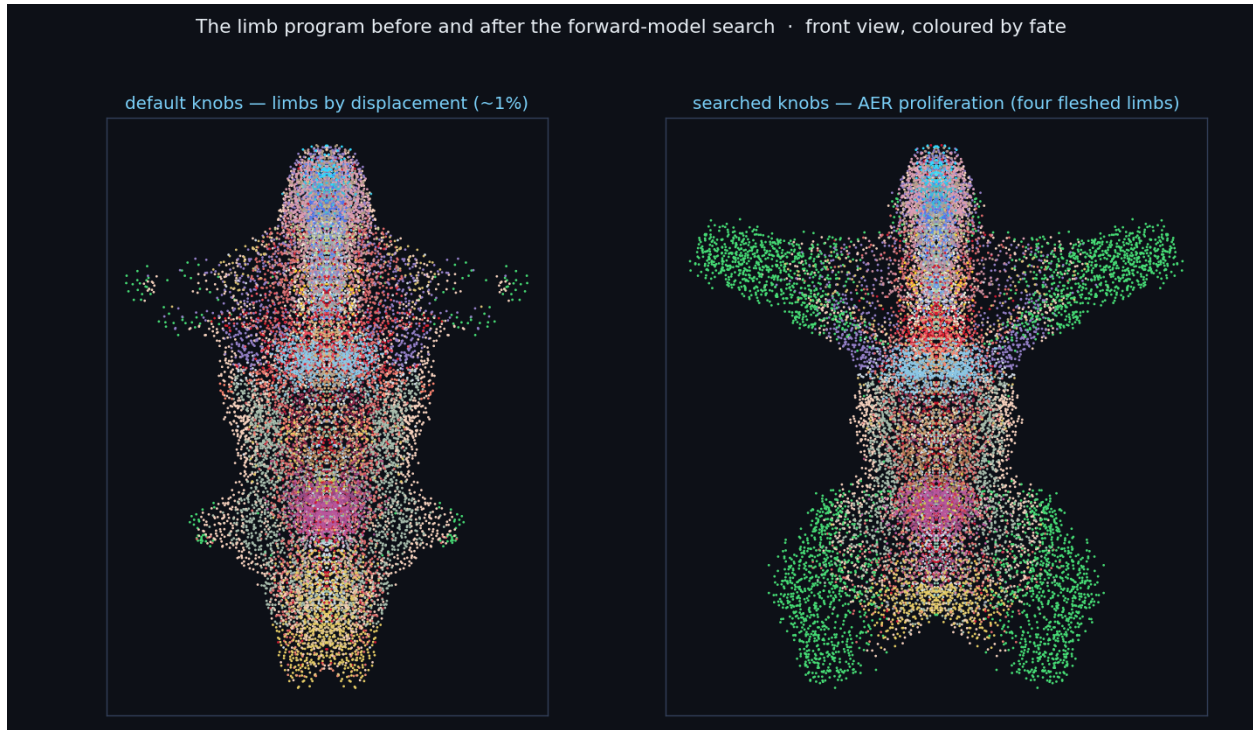


Figure 22: The limb program before and after the forward-model search, shown on the model's own cell cloud (the maturation grown from the deterministic tetrapod, not a mannequin). Left, the default knobs: the four buds are placed on the electric-body antinodes but grown only by outward displacement, so each limb is a sparse ray of about one per cent of the body's cells. Right, the searched knobs with the apical-ridge proliferation term: the limb grows by cell division to about a fifth of the body as four balanced, fleshed buds, while the axial body still populates. Faithfulness is preserved: run on a narrow (fish) body the same knobs form no buds, because the width gate that admits the left-right eigenmode is not met, so the proliferation term has nothing to amplify. Limb cells green; the axial fates in their anatomy colours.

10 The Clock Run Both Ways: from the Adult to the Blastocyst

The developmental loop has a whole-body form, and it is the single coordinate the model runs on turned end to end. Development in this framework hangs from one clock: the telomere shortens with each division and the potassium-repressive complex withdraws from the differentiation strata as it does, so a single coordinate gates when each fate unlocks, how many times the cells have divided, and how far the body has extended and folded [5]. Because the whole construction hangs from that one coordinate, it runs backward as readily as forward. And the loop of the adult and the embryo is, at the level of the whole body, this coordinate reversed.

Wound back below the developed body the coupled schedules reverse together. Fate commitment falls to zero and the committed cells return to uncommitted blastomeres as the repressive complex re-engages the embryonic archive. The antero-posterior and left-right eigenmodes of the electric frame lose their amplitude as the body ceases to be elongated, and here the reversal reaches

a result the frame itself dictates: a body carries a low eigenmode only when it is long enough to hold one, the eigenvalue scaling as the inverse square of the length, so as the axis shortens the frame collapses, the body has no axes on which to place anything, and it balls up into a radial cluster. The fetal curl un-folds and the cell number falls toward one. The radial cluster the reversal reaches is the blastocyst, and it is reached not by drawing a blastocyst but by removing the one thing that makes a body a body, its axis, so the frameless radial ball is the same frameless state the electric-body account gives for a tissue too small and too symmetric to carry a mode [3]. Measured on the reversal the body's antero-posterior-to-mediolateral aspect falls from that of an elongated tetrapod, near 2.8, to that of a ball, near 1.0, and the committed fraction from all to none, both monotone in the clock, until at the coordinate's zero a single cell remains. Run the other way the same coordinate matures the body past the fetus, its curl relaxing and its cranium-to-face proportion shifting by allometry toward the adult.

The reversal is the model's own coordinate turned backward, and its honesty is the honesty of the rest of the paper. It transforms one developed body rather than regrowing every adult organ forward, so the adult end is the allometric maturation of the grown body and the blastocyst end is the principled collapse of its frame, each carrying the tissue-conglomerate fidelity the construction works at. What it secures is that the single clock that grows the human body runs it back through the recognisable stages to the one cell it began as, so the human model the scaffold completes has a beginning as well as an end. And the developmental loop of the adult and the embryo is not only an inverse pass through the forward map for one locus but the whole body's clock read in both directions: forward the human body appears, and backward it returns to the blastocyst and the zygote.

11 Form and Function: the Anatomical and Physiological Atlases

The GWAS scaffold tests the human model against the variation between people; two older atlases test it against the invariant human it varies upon, and together they turn the single developmental loop into a triangulation. The first is the anatomical atlas, the named catalogue of every bone, muscle, vessel, and viscus assembled over more than a century of dissection, of which *Gray's Anatomy* [26] is the canonical instance; it is the *form*. Used directly as ground truth it demands that each attention head resolve its structure into its real named parts rather than an undifferentiated mass: the pectoral girdle into clavicle and scapula, the hip bone into ilium, ischium, and pubis, the presacral column into its seven cervical, twelve thoracic, and five lumbar vertebrae with a rib pair at each thoracic level, the sternum into manubrium, body, and xiphoid, the skull into its neurocranial and facial bones. On the same frame the recognisable features that make a body read as a person are placed not by hand but by the recursion the design paper gives one scale below the body: the facial prominences, and the orbits, nose, cheeks, and chin they carry, sit on the antinodes of the face's own gap-junction eigenmodes, and the limb carries its own such frame, its proximo-distal, antero-posterior, and dorso-ventral axes the low eigenmodes of the limb cells' operator and its five digits the antinodes of the antero-posterior mode, the same *Shh*-patterned wavelength that sets digit number. The individual adapter then writes the GWAS effect sizes onto these features as displacements from the neutral form, a per-allele regression coefficient becoming a positional

delta. So the named anatomy is both carved by the kernel and varied by the scaffold.

The second atlas is physiological. Where the anatomical atlas gives form, the Physiome Project [27] and, for developmental bioelectricity, the bioelectric tissue simulation engine BETSE [28] give *function*: a curated electrophysiological model for each tissue. Their governing equation already contains the decomposition this series relies on, for a tissue model is at bottom the monodomain cable equation $\partial_t V = \nabla \cdot (D \nabla V) + I_{\text{ion}}(V, \mathbf{g})$, whose diffusion term $\nabla \cdot (D \nabla V)$ is the gap-junction operator that furnishes the electric-body, electric-face, and electric-limb eigenframes, and whose reaction term I_{ion} is the local rule the differentiation and channel heads occupy. What these mature simulators lack is not the physics but the two learned layers that make the physics a compiler: no map from the genome to the parameters, so each conductance is fitted by hand to voltage-clamp data one channel at a time rather than emitted from sequence by a genomic model, and no learned differentiable local rule in place of the hand-derived ionic term. The present construction is therefore complementary to that mechanistic tradition and contributes the genome-connected, learned layer it never had. And the two meet in practice, because the same gap-junction operators that place each organ's subheads on its own eigenframe antinodes—the oral–aboral axis of the gut, the cortico-medullary axis of the kidney, the optic axis of the eye—can be weighted directly by the real per-tissue conductances of the Physiome models, the fast ventricular and slow nodal cardiac coupling, the interstitial slow-wave coupling of the gut, so that each head carries its operator from the physiological atlas as it carries its roster from the anatomical one.

The three atlases make the method a triangulation. Where the construction began as genome $\leftrightarrow$ heads $\leftrightarrow$ development atlas, each attention head is now bound to four texts at once: to the genome through its master-transcription-factor super-enhancer cluster, to development through its spatial-transcriptomic annotation, to form through its named structure in the anatomical atlas, and to function through the curated model of its tissue. The binding is countable: of twenty-five organ heads and subheads all twenty-five are anchored on the genome and on the developmental atlas, eighteen carry a named anatomical structure, sixteen carry a canonical physiological model, and sixteen are grounded on all four at once—the cardiac ventricle, for instance, reading its identity from *Irx4/Hand2*, its territory from the heart annotation, its chamber from the anatomical roster, and its conduction from a human ventricular action-potential model. The force of it is that these are independent assays that do not check one another but each check the head from a different side: whether it is built of the right cells, resolves into the right named parts, and reproduces the right physiology. Two honesties bound the claim. The anatomical and physiological atlases are complementary rather than mutually verifiable, giving form and function respectively, so their agreement is agreement about the head and not about each other. And only the subset of the physiological repository that corresponds to a head's tissue is a model of the head itself, the remainder being subcellular or pathway models that would ground a head's internal machinery rather than the head. Within those bounds the human the scaffold completes is auditable, at once, against how each named part is specified in the genome, where it sits in the embryo, what it is called in the anatomy, and how it behaves in the physiology.

The audit is not only countable but visible on the model's own matured body (Figure 23). Two of its faces are drawn directly from the anatomical atlas. The first is the named surface musculature: the deltoid, pectoralis, the rectus abdominis with its tendinous six-pack, the trapezius and latissimus of the back and the gluteus and gastrocnemius of the limb are each a fusiform relief the skin reads over, so the body reads as a nameable figure rather than a smooth envelope, and each swell is a lobe generated by the one soft-tissue mechanism rather than sculpted by hand. The

second is the head-to-toe placement of the viscera. Because organ position is a developmental invariant and not a polygenic quantity—the boundary the association catalog itself drew above—each organ is carried to its Hox-addressed axial level by a single antero-posterior placement head that translates a condensed organ as a body while leaving every other cell untouched, the generalisation of the cardiac descent. Scored against the canonical anatomical levels, the mean placement error falls from about a seventh of stature to near zero and the twenty-two head-to-toe order inversions of the un-placed body—the heart sunk onto the kidneys, the adrenal below the kidney, the liver below the pancreas—all resolve. So the heart sits in the thorax above the liver, the liver above the paired kidneys, and the bladder in the pelvis, each organ on the level *Gray's* names for it.

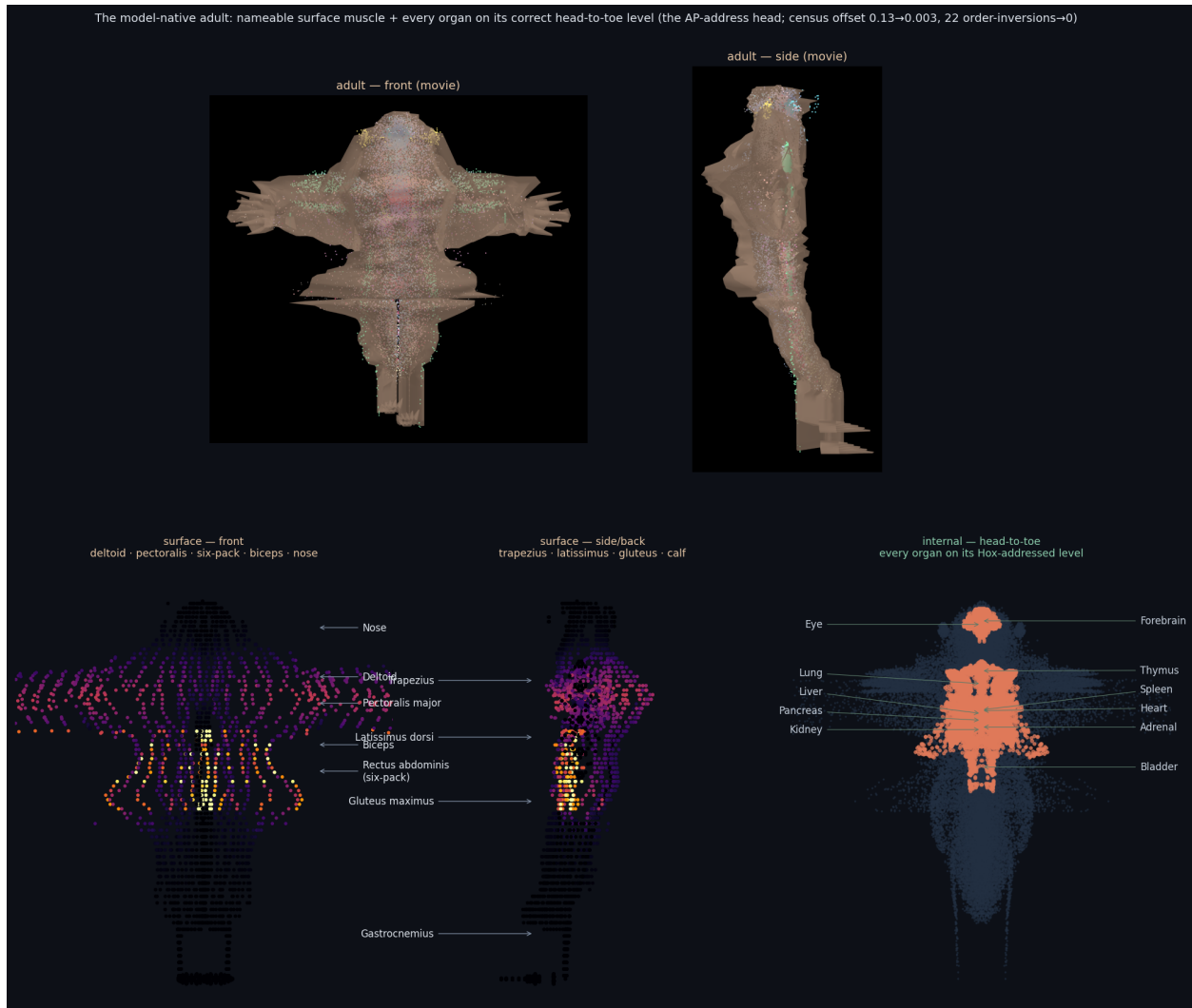


Figure 23: The model’s own matured adult, audited against the anatomical atlas and found nameable. **Left, the movie’s own adult:** the viewer’s render of the standing adult, front and side (the same movie as Figure 5). **Centre:** the surface musculature that reads through the skin, front and back—the deltoid, pectoralis, and rectus six-pack in front, the trapezius, latissimus, gluteus, and gastrocnemius behind—each a fusiform relief generated by the single soft-tissue mechanism and coloured by its height above the envelope. **Right:** the viscera, every organ carried to its Hox-addressed head-to-toe level by the antero-posterior placement head (the generalisation of the cardiac descent, a condensed organ translated as a body, read-only on the rest). Scored against the canonical anatomical levels, the mean placement error falls from about 0.13 of stature to 0.003 and the twenty-two head-to-toe order inversions of the un-placed body all resolve: the heart sits above the liver, the liver above the paired kidneys, the bladder in the pelvis, each on the level the anatomy names.

The whole adult carries the same nameable anatomy, and the parts that had read as stumps now read as a body (Figures 3 and 24). The skin surface, built from the model’s own outer cells and shaded by the muscle relief, closes over the deltoid, the pectoral and the rectus six-pack; each

arm ends in a five-fingered hand grown as the five *Shh*-patterned rays of the limb, and each leg in a five-toed foot; and the head carries the face. These are not appliquéd onto a template but the model's own tissue, the fingers and toes the same digit program the limb chondrogenesis head runs one scale below the body. So the figure that ends the developmental movie is the same cell cloud that began it as one cell, now fleshed and nameable head to toe.

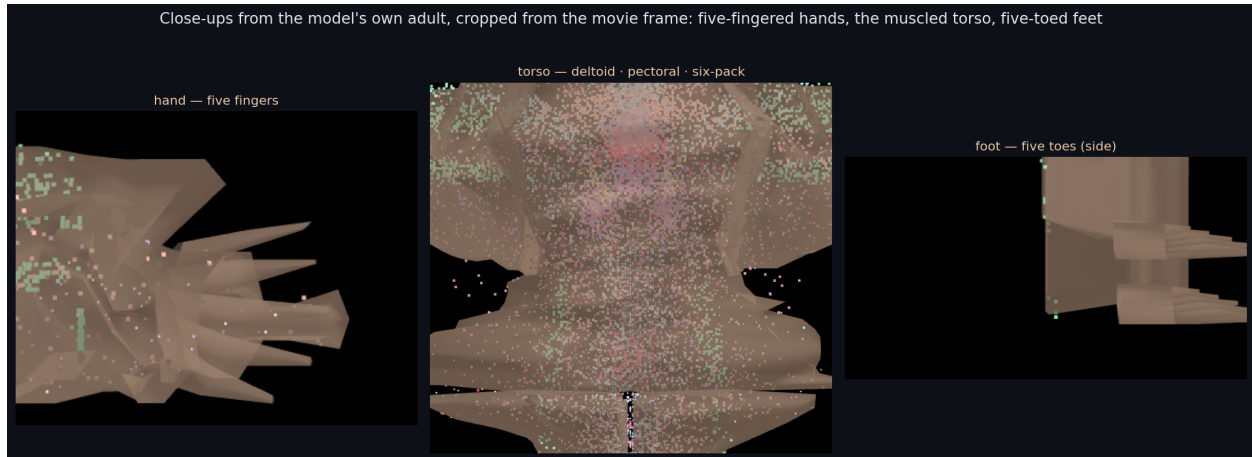


Figure 24: Close-ups from the model's own adult: the parts that make it read as a body. A five-fingered hand (the five *Shh*-patterned digit rays grown from the autopod), a five-toed foot in side view, the muscled torso with its deltoid, pectoral and rectus six-pack relief, and the whole fleshed figure. All are the model's own skin surface, shaded by the muscle relief; no scanned mesh is used.

12 The Economy of the Compiler: Many Metrics, Few Mechanisms

The three atlases raise a question the construction must answer plainly, because the anatomical atlas can be descended without end: from the body to its organs, from an organ to its subheads, from the heart to its four chambers and from a chamber to the layers of its wall, from a tissue to its cell types. If closing the distance to *Gray's* demanded a metric for every named structure at every scale, and behind each metric a mechanism to generate it, the model would grow in complexity as fast as the catalogue it is checked against, and having reproduced one human it would have learned nothing that built a second. The central economy of the method is that it does not. The number of *metrics*—the named structures the atlases enumerate and against which the model is scored—is large and grows with the depth of resolution one demands. The number of *mechanisms*—the distinct generative rules from which those structures are built—is small, and does not grow with it.

The reason is that the mechanisms are reusable primitives applied recursively down the hierarchy, not bespoke rules laid one per part. The primitives are of order fifteen—mesenchymal condensation, lateral-inhibition spacing, placement on the antinodes of a gap-junction eigenframe, epithelial tube branching, connective-tissue cleavage, chondrogenic segmentation, the clock-and-wavefront, directed migration, convergent extension, apical constriction, cadherin sorting, and integrin adhesion—and each is invoked wherever its field recurs, at whatever scale. The same lateral-

inhibition operator that spaces the teeth along the dental lamina spaces the valves of the heart, the placodes of the skin, and the digits of the hand at the Shh-patterned wavelength. The same eigenmode-antinode placement that seats the heart at its antero-posterior level within the body seats the chambers at their levels within the heart, one scale down, on the heart's own gap-junction operator. The same connective-tissue cleavage that carves the axial myotome into its named muscles carves the limb mass into biceps and triceps. A structure at a finer scale is therefore not a new mechanism but the same mechanism reading a finer field, and the recursion the design paper states for the face and the limb—the sub-parts on the antinodes of the part's own operator—is the general form of it. Adding a metric for the heart's chambers, or its valves, or the layers of its wall, adds an audit; it does not add a generator.

The counting makes the economy explicit (Table 4). A recognisable human, resolved to its named bones, muscles, viscera, and the features of its face, is a target of order a thousand metrics—two hundred-odd bones, some six hundred muscles with their origins and insertions, a few score organs each fanning into its subheads, a few hundred cell types—and a complete descent to every named foramen and tissue layer multiplies that again. The mechanisms that generate all of them are of order fifteen reusable primitives. Instantiated as one head per locus each primitive serves they become of order fifty to a hundred heads, of which the twenty-five organ heads and subheads of the previous section are the visible core. And the independent degrees of freedom—the dimensions along which one body differs from another, the coupled head manifold the mouse paper measures at order seven, upon which the species and individual adapters move—are fewer still. The counts stand in a fixed order: of order a thousand metrics, generated by of order fifteen reusable primitives (of order a hundred once instantiated per locus), moving on of order ten degrees of freedom.

That ordering is the signature of a compiler as against a catalogue. A lookup table of the human would need a parameter for every metric, would grow without bound as the atlas is descended, and having memorised one human could generate no other. A compiler needs a rule for every mechanism, and the mechanisms being few and reusable it closes the distance to the atlas at a cost that saturates, while the same rules under a different low-rank adapter grow a different human—a woman, a child, a carrier of a variant—and so stay falsifiable and predictive where the table is neither. The genome sets the ceiling on the count directly, for each distinct structure is specified by a master-transcription-factor super-enhancer head and there are of order a few hundred such heads in the regulatory genome; the thousands of metrics fan out from those few hundred heads by the recursion, and the tens of degrees of freedom are the low-rank manifold on which the heads co-vary. Getting closer to *Gray's*, then, is the work of adding mechanisms until the recognisable human is complete and then descending by the same reusable rules—not of adding parameters without end. The atlas is the objective the compiler is scored against, and the compiler stays small while the objective grows.

Table 4: The economy of the compiler: the validation surface grows with resolution, the generator does not. Orders of magnitude, for a recognisable human resolved to its named parts.

Level	Order	What it is
Metrics (validation)	10^3 – 10^4	named structures $\times$ properties: bones, muscles with origin and insertion, organs, subheads, cell types, at every scale of the atlas
Mechanisms (primitives)	~ 15	reusable generative rules: condensation, lateral inhibition, eigenframe-antinode placement, tube branching, connective-tissue cleavage, chondrogenic segmentation, clock–wavefront, migration, convergent extension, apical constriction
Attention-head instantiations	~ 50 – 100	primitive $\times$ locus; the twenty-five organ heads and subheads its visible core
Genome attention heads (ceiling)	$\text{few} \times 10^2$	master-transcription-factor super-enhancer clusters; the structural count the genome carries
Degrees of freedom	~ 7 – 20	the coupled head manifold (order seven, per the mouse paper), with the maturation and pigmentation axes; the dimensions along which humans differ (species and individual adapters)

13 Scope

The construction is stated as an architecture with a seed for each layer, not as a finished human model. The kernel and the human-embryo transfer are measured; the individual adapter and the three-way manifold test are laid out on the design paper’s foundation. The developmental forward map is the open layer, and the developmental loop is a principle with a one-organ programme, not yet a fitted result across the body. What is claimed is that the four layers compose, that the adult end can supervise the embryonic end through the forward map, and that the heart is where the loop can first be closed against data.

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